Hands-on Introduction to NGS Data Analysis

Lecturers: Lucrezia Valeriani[1,2], Giorgia Gandolfi[2,3]

[1]AREA Science Park, [2]University of Trieste, [3]IRCCS Ospedale San Raffaele

Organisers:
Giulio Caravagna, CDSLab - University of Trieste
Stefano Cozzini, RIT - AREA Science Park

Day 1 - 10 November 2026



Cancer Data Science (CDS) Lab

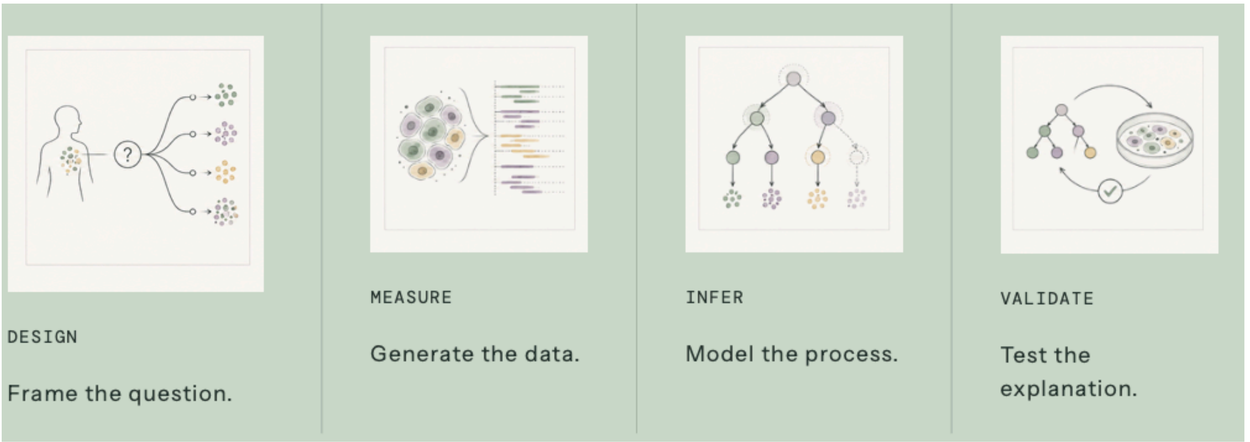
The lab studies how **tumours** develop, adapt and respond to treatment, creating new ways to **interpret** complex biomedical data.

Develops **model-based machine learning** for bulk, longitudinal, single-cell, spatial and long-read assays, connecting experimental design, **uncertainty-aware** inference and biological validation.

Interdisciplinary Background:

Statistics, Physics, Computer Science,
Biology, Data Science etc.

Collaborations with hospital and research institutions:

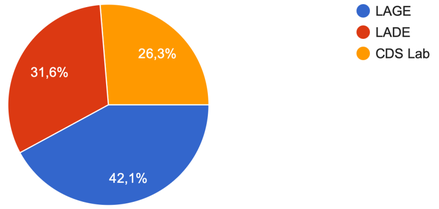


Overview of the participant

Who's in the room?

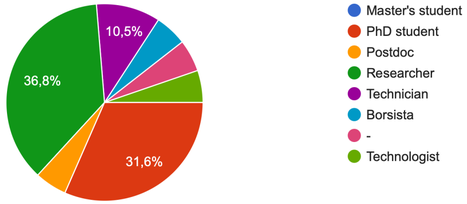
Which is your research group?

19 risposte



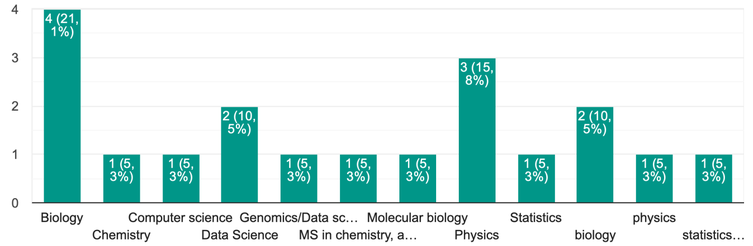
What is your current position?

19 risposte



What's your background (biology, physics, etc.)

19 risposte



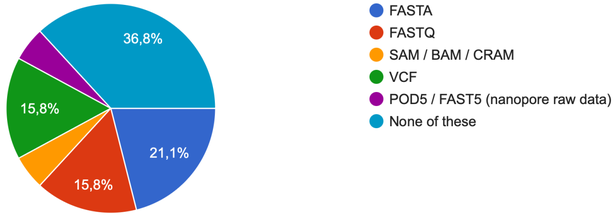
Have you ever run a workflow manager (e.g. Nextflow, Snakemake)?

19 risposte



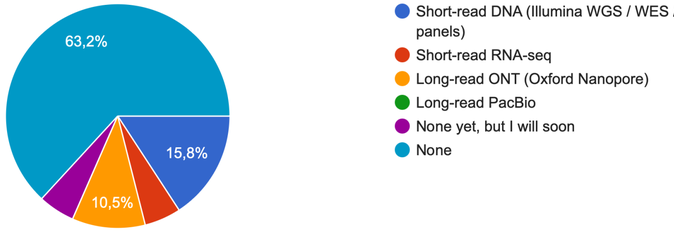
Which file formats have you worked with directly

19 risposte



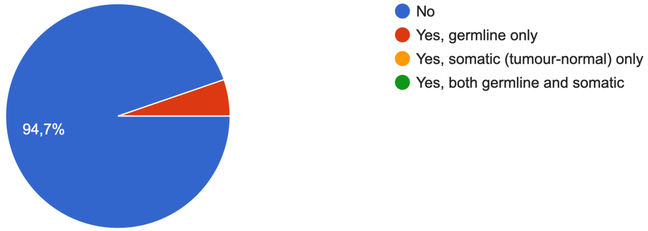
Which sequencing data have you analysed yourself?

19 risposte



Have you done variant calling before?

19 risposte



Overview of the course

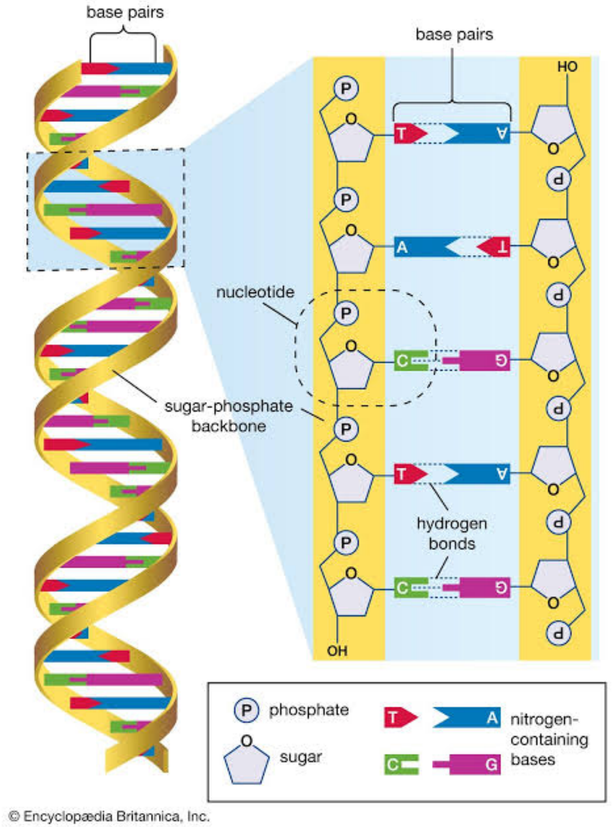
The two-day plan

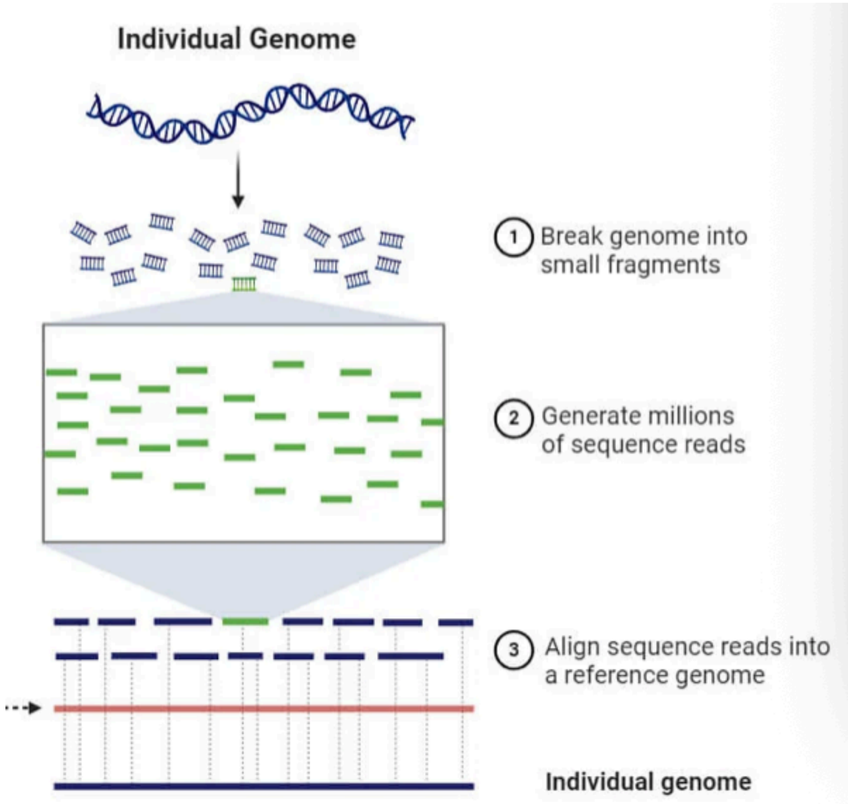
Session	Topics
Day 1 - Theory (3h)	NGS fundamentals, short vs long reads, file formats (FASTQ, FASTA, BAM, VCF), Phred scores, demultiplexing, alignment, germline vs somatic variant calling, GATK Best Practices and nf-core/sarek
Day 1 - Practice (1h)	Exploring FASTQ files and Phred scores, launching Sarek, read QC and trimming within Sarek
Day 2- Short-read practice (1.5h)	Alignment QC and preprocessing (duplicate marking, BQSR), germline & somatic calling, reading VCFs
Day 3 - Long-read practice (2.5h)	Basecalling, long-read QC, alignment, variant calling with the EPI2ME pipeline, short vs long comparison

Course material: https://github.com/caravagnalab/ngs_course/tree/main

DNA and Whole Genome Sequencing

What and how can we read from DNA?



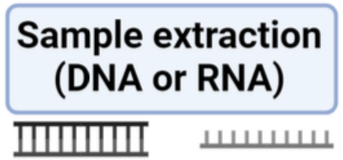


https://microbenotes.com/whole-genome-sequencing/

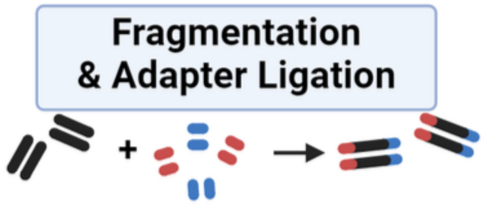
NGS: Next-Generation Sequencing Workflow

How Illumina turns DNA into reads

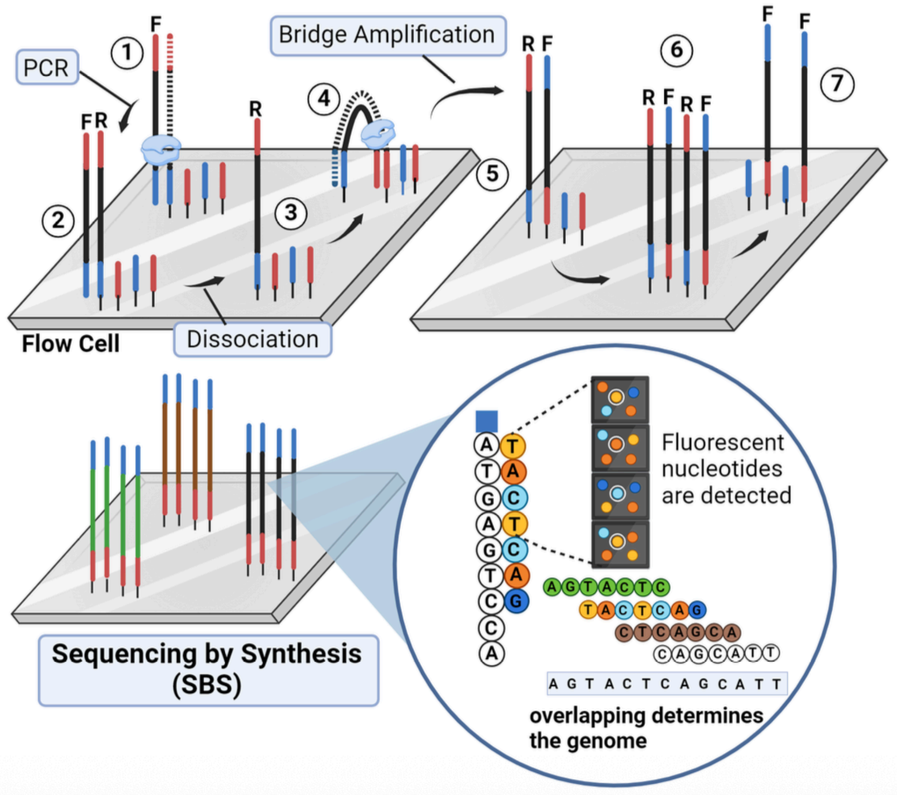
1. Sample Preparation



2. Library Preparation



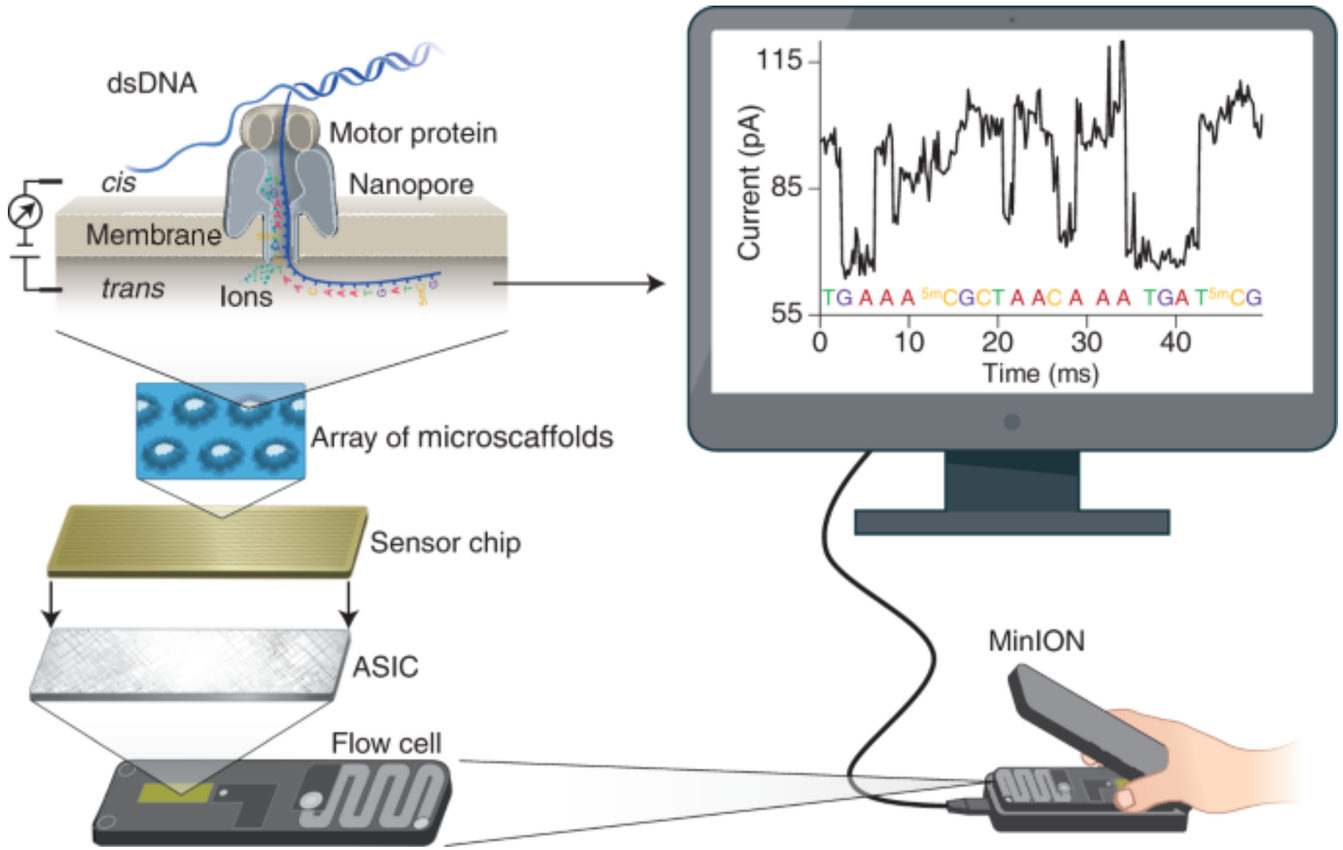
3. Sequencing



4. Basecalling

Third Generation Sequencing

Oxford Nanopore long-read sequencing



Wang, Yunhao, et al. "Nanopore sequencing technology, bioinformatics and applications." *Nature biotechnology* 39.11 (2021): 1348-1365.

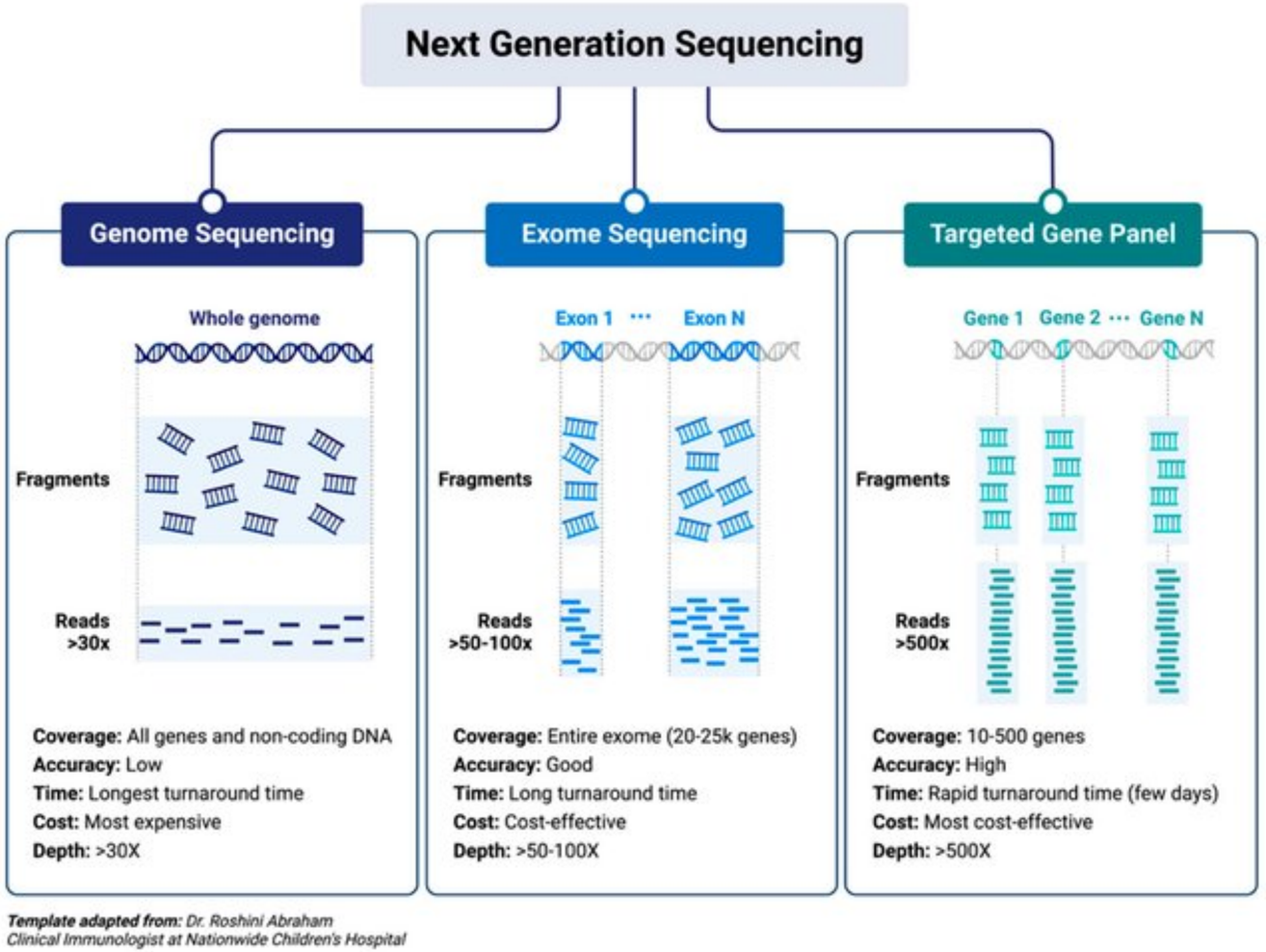
Short-read vs long-read sequencing

The choice of sequencing technology shapes what biological questions you can answer.

	Short-read (Illumina)	Long-read (PacBio / ONT)
Read length	150–300 bp	1 kb – 100+ kb
Accuracy	Very high (Q30+)	Lower raw accuracy, but improving rapidly
Throughput	Very high	Lower, though increasing with newer platforms
Cost per Gb	Low	Higher, dropping fast
SNV / indel calling	Excellent — the gold standard	Good
SVs & CNVs	Limited - short reads struggle to span breakpoints	Excellent - reads cross the entire variant
Repetitive regions	Poor - reads too short to map uniquely	Excellent - reads span through repeats
Methylation	Requires bisulfite conversion (extra step)	Direct detection, no conversion needed
Typical use cases	WGS, WES, gene panels, somatic calling	SV detection, de novo assembly, phasing, methylation

Different Types of Sequencing

Whole genome, exome, or panel?



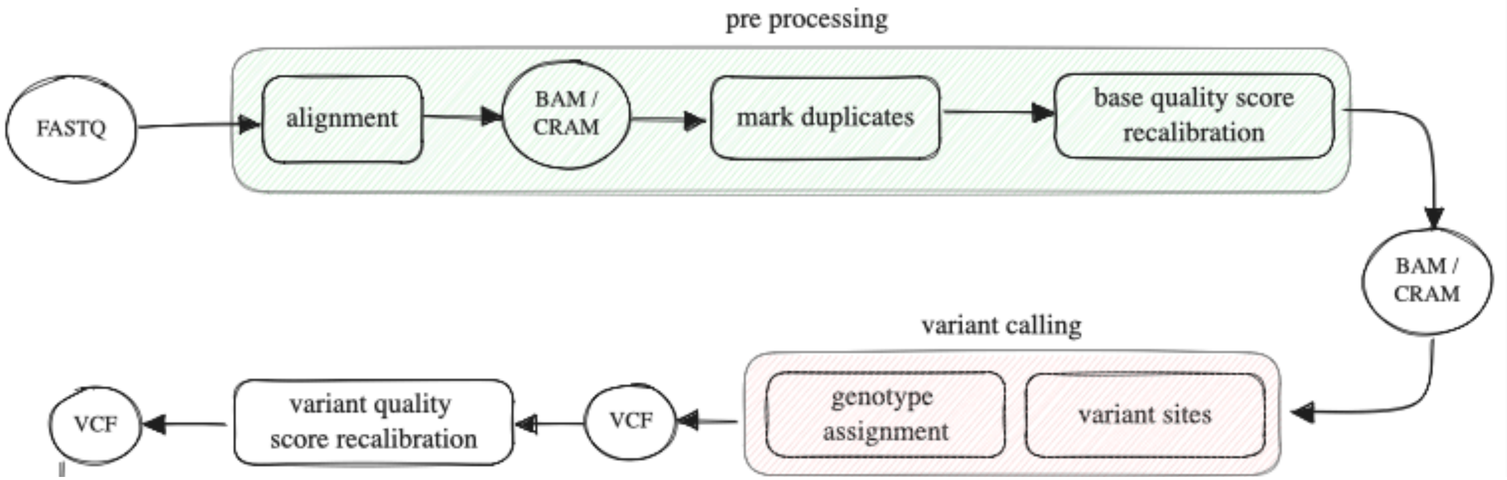
Template adapted from: *Dr. Roshini Abraham*
Clinical Immunologist at Nationwide Children's Hospital

Mohamed, Wael. "Equity in neurogenomics: advancing Parkinson's disease research in AfrAbia." *Academia Molecular Biology and Genomics* 2.2 (2025).

Data Analysis Workflow

From reads to variant calls - GATK standards

1. Process raw sequencing data
2. Align pre-processed reads to the reference genome
3. Post-process aligned reads
4. Identify variants of interest.



Reference Genome: FASTA

The reference provides the sequence and coordinates used to place reads.



- A reference genome is a collection of contigs.
- A contig is a stretch of DNA sequence

GRCh37/38 (maintained by the NCBI/Ensembl) typically uses plain numbers (1, 2, X).
hg19/hg38 (packaged by the UCSC Genome Browser) prefixes chromosome names with chr (chr1, chr2, chrX).

Tag	Value	Meaning
chr1	—	**Sequence name** — the primary identifier, here chromosome 1
AC	CM000663.2	**Accession number** — the versioned GenBank/INSDC accession for this sequence
gi	568336023	**GI number** — a legacy NCBI numeric identifier (largely deprecated now)
LN	248956422	**Length** — the sequence is 248,956,422 bases long
rl	Chromosome	**Role/type** — this sequence represents a full chromosome (vs. scaffold, patch, etc.)
M5	6aef897c3d6ff0c78aff06ac1	**MD5 checksum** — a hash of the sequence itself, used to verify integrity and match sequences across files
AS	GRCh38	**Assembly** — this sequence belongs to the GRCh38 human reference genome

```
>chr1 AC:CM000663.2 gi:568336023 LN:248956422 rl:Chromosome
M5:6aef897c3d6ff0c78aff06ac189178dd AS:GRCh38
NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
NNNNNNNNNNTATTCAAAAAATTGAGAATTTCTGACCACTTAACAAACCCACAGAAAATCCA
CCCGAGTGCACTGAGCACGCCAGAAATCAGGTGGCCTCAAAGAGCTGCTCCCACCTGAA
GGAGACGCGCTGCTGCTGCTGTCGTCCTGCCTGGCGCCTTGGCCTACAGGGGCCGCG
GTTGAGGGTGGGAGTGGGGGTGCACTGGCCAGCACCTCAGGAGCTGGGGGTGGTGGT
GGGGGCGGTGGGGGTGGTGTTAGTACCCCATCTTGTAGGTCTGAAACACAAAGTGTGGG
GTGTCTAGGGAAGAAG
…
>chr2 AC:CM000664.2 gi:568336022 LN:242193529 rl:Chromosome
M5:f98db672eb0993dcfdabafe2a882905c AS:GRCh38
NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
NNNNNNNNNAAAAGCATTTATGCTACAAATTACTATGGTAATTATGCTACAAATTTATGGTAC
CATAAATTACCATAGTAATTTGTAGCATAAATTTGTACTATGGTACAAATTACATGGGAGAGT
GAAGGTGGGTTAAAACATTCATATTAAAGAACTTCCACTCAGATTGCAAGAAAAGAGAGAG
AATGGAGATGGTAGCACAAGTCCCTACAATAAAAGTAGATGTTTTGAGATCAGTTCTATTT
```

Raw Sequencing Data: FASTQ

Anatomy of a read



A FASTQ record stores a sequence and one quality character per base.

Read

Read identifier

Nucleotide sequence

Separator (+)

Base quality (ASCII code)

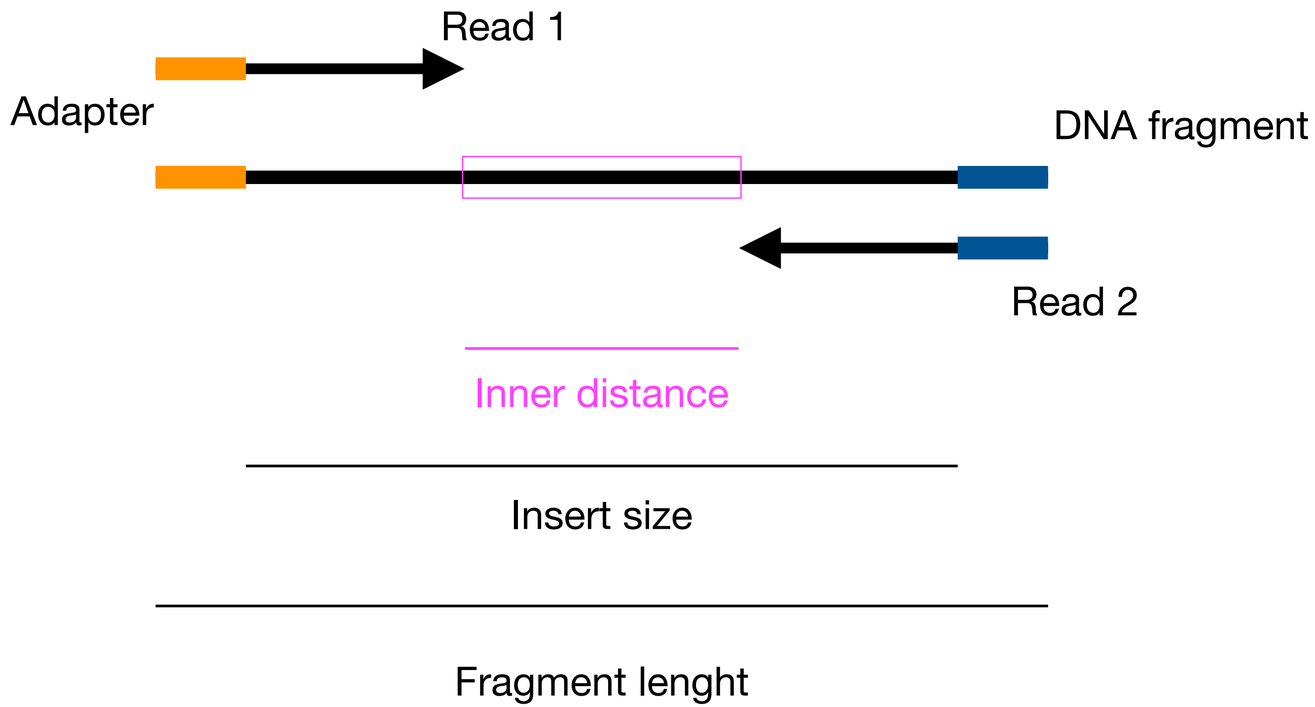
```
@A00410:214:HKG7JDRXX:1:2101:1750:1016 1:N:0:CTCACAAC+AAGGAGAC
TNCTGAGGCAGAATTTCGAGATGAAGGTCTTGTATAATTTTCAGCATACTGTTTGTCAGATTTTTCATCCAACTCTTTTTGTTGTCGTTCCAGTTCTCTC
+
F#FFFFFFFFF:FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF:FFFFFFFFFFFFFFF:FFFFFFFFFFFFF
@A00410:214:HKG7JDRXX:1:2101:2546:1016 1:N:0:CTCACAAC+AAGGAGAC
CNTGGATCTTGGCCTTTACATTTTCTATCGTATCCGAGGGTTCAACCTCGAGGGTGATGGTCTTCCCCGTAAGGGTTTTCACGAAAATCTGCATTTTGGT
+
F#FF:FFFFFFFFFFFF:FFFFFFFFFFFFFFFF,FFF:F:FFFFFFF,FFF,:FFFFFFFF:FFFFFFFF::FFF,FFFF,F:FF,::,FF,:F:F,F
@A00410:214:HKG7JDRXX:1:2101:3215:1016 1:N:0:CTCACAAC+AAGGAGAC
GNTAAGTCCTATGTCTTTTGTGAAATTCCAATACTTAAACTGCAAGTCTGCAATCGTCTCTGAAGTCAGTGAAATTAAGAAAA
+
##(9B?AFFFFFFFFFFFFFFFFFFFFFFFFFFFF:FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
```

Field	Meaning
Instrument name	Sequencer ID
Run/flowcell ID	Which sequencing run
Lane, tile, x, y	Physical position on the flowcell
/1 or /2	Read 1 or Read 2 (paired-end)

@instrument:run:flowcell:lane:tile:x:y read:filtered:controlbits:index

FASTQ: Paired-end sequencing

What are R1 and R2?



sample_name_R1.fastq.gz

@A00618:447:HV22YDMXY:1:1104:8323:30718
1:N:0:TAATTAGCGT+TTCTTAACCA

GGTCATTTGTGGGCATCAGTTCATAATGTACAAAATAAAGTTTATGAAAA
AGTTAAATCTGAAAAAGTTGGAATCACCTTAGTGCAAGCACCTACTAAA
TAATAAGAGAATGGTTAAGAGTCATTTCTTGAAGTTTATTCCTGAGGTCT
TT

+

:FFFFFFFF:F,:FFFFFFFFFFFFF:FFF:FFFF:FFF::FFFF:FFFFFFFFFFFF,:FF
F:FF,FFF:FFFFF,:FFF:F:,F,F,FF,FFF:FFFFFFFFF:FFFFFF,FFFFFFFF:FF:F:F
F:FFFFFFFFFFF,FF,FF:FF

sample_name_R2.fastq.gz

@A00618:447:HV22YDMXY:1:1104:8323:30718
2:N:0:TAATTAGCGT+TTCTTAACCA

GTGTATACCCTGCCTACATGGCAAGCAGTCCTCTGTCTGTTCAGCATC
TCACCATTTTCCAGATTTATATTGGAGACATCCAGTGATGGACTGCATTT
CCCTATGATTTCTCTTTCCTAATAAGCTTGTAGCCAAGTTCCAATAATATA
TT

+

FFFFFFF:FFF:FF:FFFFFFFFFFFFFF:FFFFFFFFFFFFFFFFFFF:F,FFFFFFF
FFFFFFFFFFFFFFFFFF:FFFFFFFFFF:FFFFFFFFFF,FF:FFF:FFFF,:FF:FF,F
FFFF:FFFFF:FFFFFFFFFF:FFF:F:FF,

FASTQ: Quality Score

What does Q30 mean?

The quality score of a base, also known as a **Phred** or **Q** score, is an integer value representing the estimated probability of an error, i.e. that the base is incorrect.

If P is the error probability, then:

$$\text{Phred quality score: } Q = -10 \log_{10} P$$

$$\text{Base error probability: } P = 10^{\frac{-Q}{10}}$$

Quality	Probability of error	Accuracy
10 (Q10)	1 in 10	90%
20 (Q20)	1 in 100	99%
30 (Q30)	1 in 1000	99.9%
40 (Q40)	1 in 10000	99.99%

ASCII_BASE=33 Illumina, Ion Torrent, PacBio and Sanger

Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII
0	1.00000	33 !	11	0.07943	44 ,	22	0.00631	55 7	33	0.00050	66 B
1	0.79433	34 "	12	0.06310	45 -	23	0.00501	56 8	34	0.00040	67 C
2	0.63096	35 #	13	0.05012	46 .	24	0.00398	57 9	35	0.00032	68 D
3	0.50119	36 $	14	0.03981	47 /	25	0.00316	58 :	36	0.00025	69 E
4	0.39811	37 %	15	0.03162	48 0	26	0.00251	59 ;	37	0.00020	70 F
5	0.31623	38 &	16	0.02512	49 1	27	0.00200	60 <	38	0.00016	71 G
6	0.25119	39 '	17	0.01995	50 2	28	0.00158	61 =	39	0.00013	72 H
7	0.19953	40 (	18	0.01585	51 3	29	0.00126	62 >	40	0.00010	73 I
8	0.15849	41)	19	0.01259	52 4	30	0.00100	63 ?	41	0.00008	74 J
9	0.12589	42 *	20	0.01000	53 5	31	0.00079	64 @	42	0.00006	75 K
10	0.10000	43 +	21	0.00794	54 6	32	0.00063	65 A			

sample_name_R2.fastq.gz

@A00618:447:HV22YDMXY:1:1104:8323:30718 2:N:0:TAATTAGCGT+TTCTTAACCA

GTGTATACCCTGCCTACATGGCAAGCAGTCCTCTGTCTGTTCAGCATCTCACCATTTTCCAGAT
TTATATTGGAGACATCCAGTGATGGACTGCATTTCCCTATGATTTCTCTTTCCTAATAAGCTTGTA
GCCAAGTTCCAATAATATATT

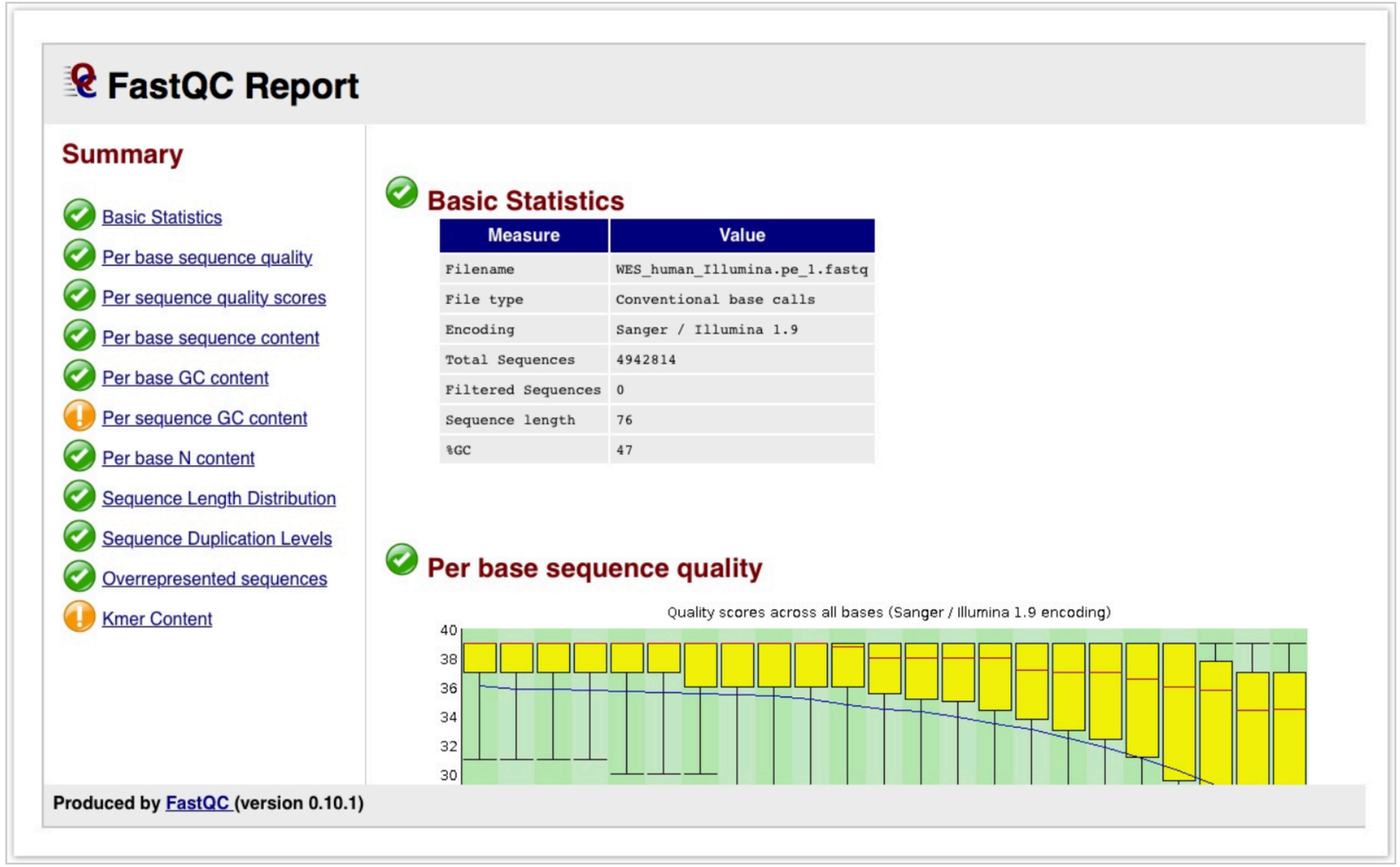
+

FFFFFFF:FFF:FF:FFFFFFFFFFFFFF:FFFFFFFFFFFFFFFFFFF:F,FFFFFFFFFFFFFFFFFFFFFFFF
F:FFFFFFFFFF:FFFFFFFFFF,FF:FFF:FFFF,:FF:FF,FFFFF:FFFFF:FFFFFFFFFF:FFF:F:FF,

Quality Control: FASTQC

What FastQC tells us

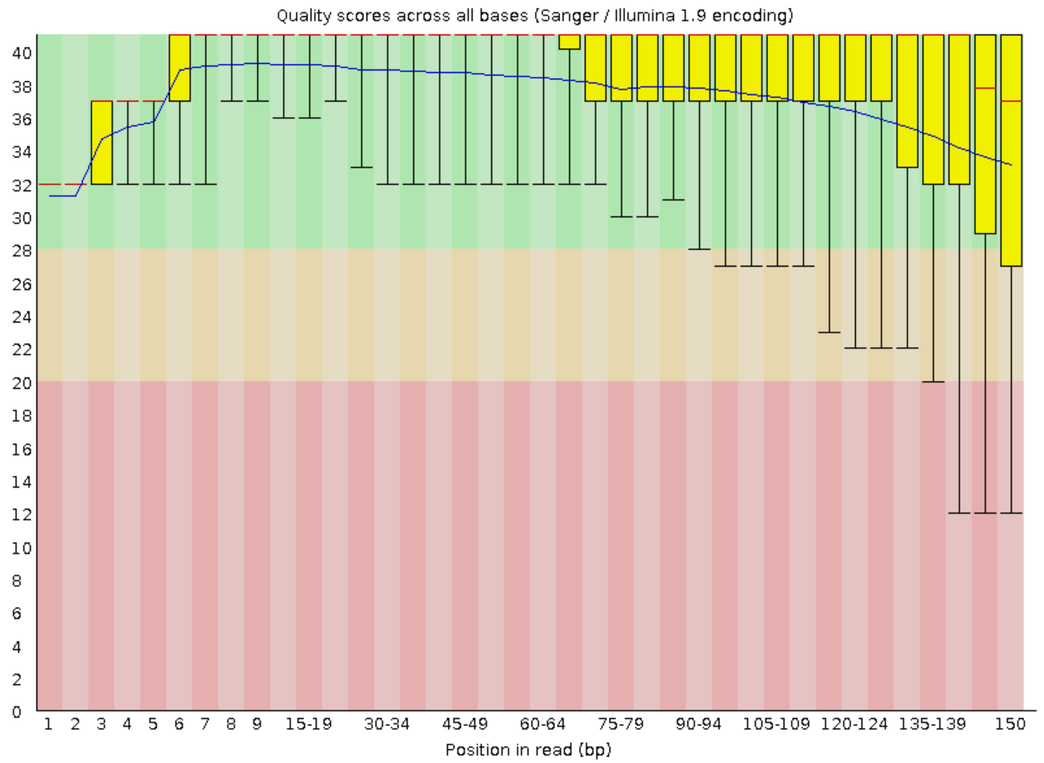
A QC report flags patterns worth investigating before alignment.

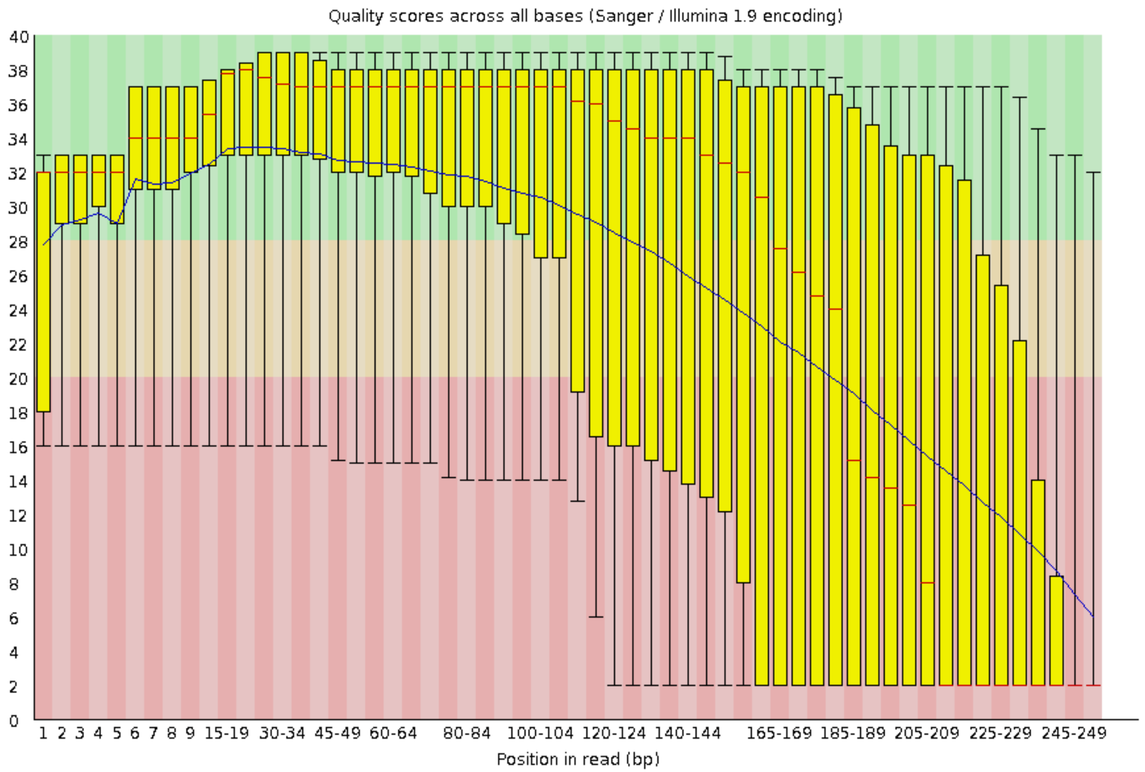


https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

Quality Control

Per base quality





https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

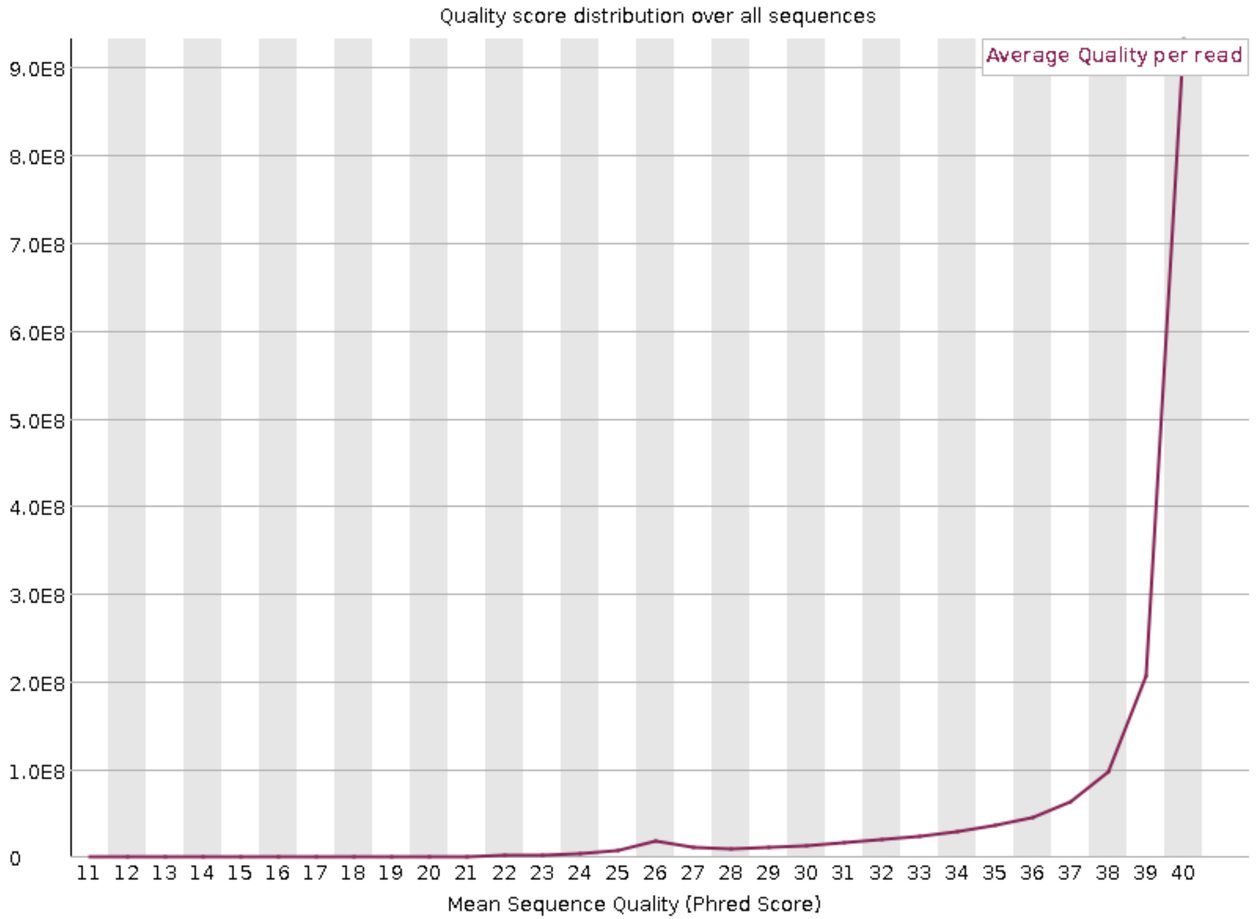
Quality Control

Per sequence quality control

The per sequence quality score report allows you to see if a subset of your sequences have universally low quality values.

It is often the case that a subset of sequences will have universally poor quality, often because they are poorly imaged (on the edge of the field of view etc), however these should represent only a small percentage of the total sequences.

If a significant proportion of the sequences in a run have overall low quality then this could indicate some kind of systematic problem - possibly with just part of the run (for example one end of a flowcell).



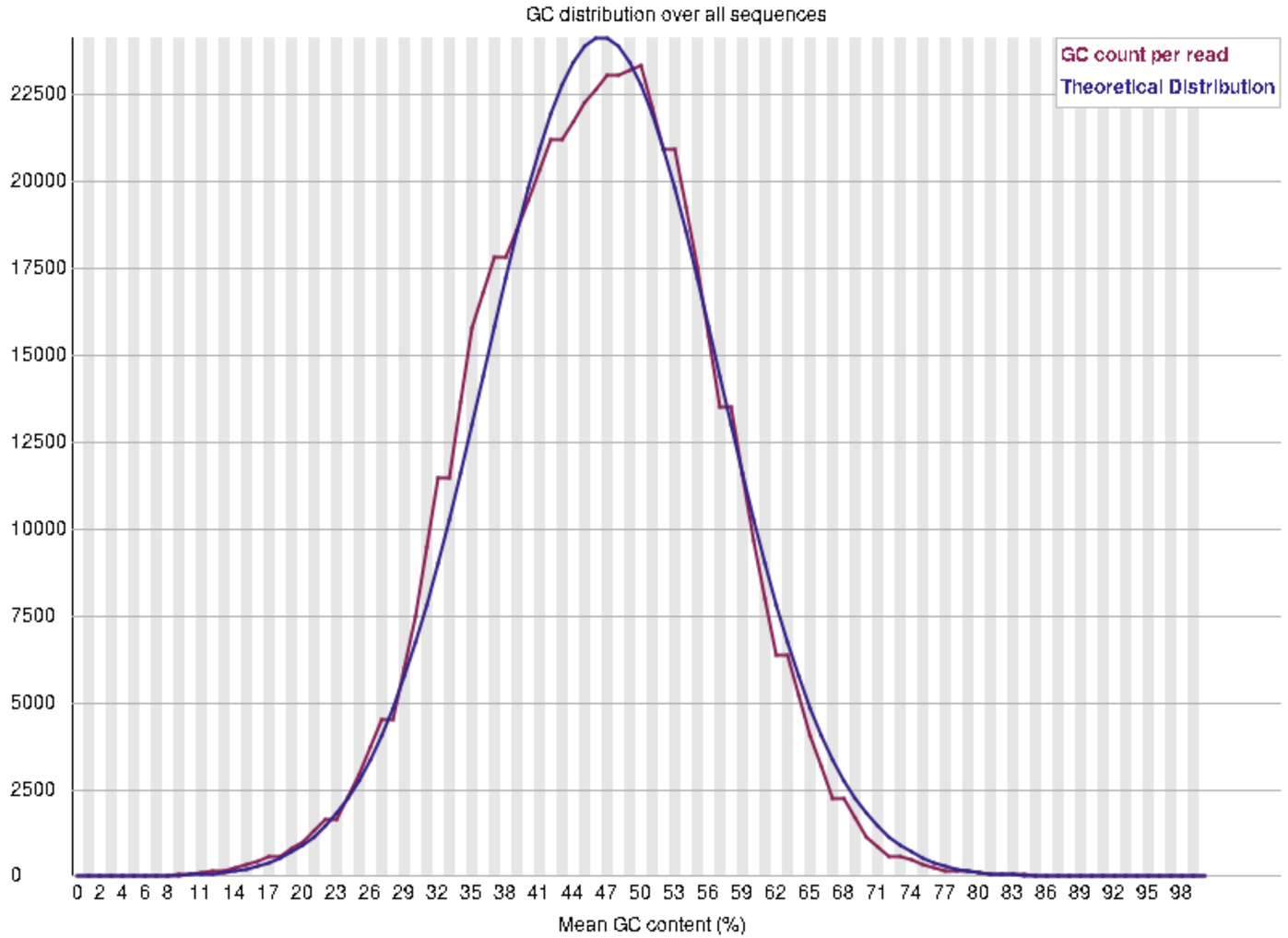
https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

Quality Control: FASTQ

GC content

GC- and AT-rich regions are more difficult to amplify

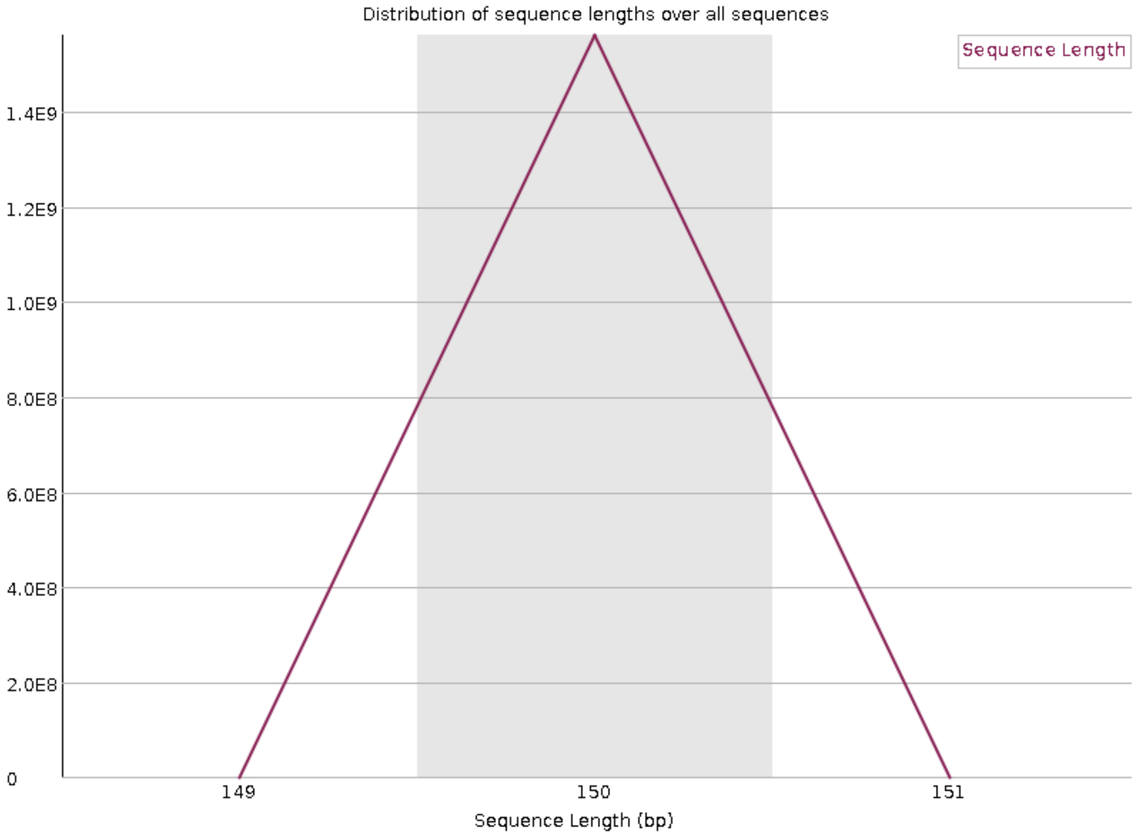
Compare the GC content against the expected distribution (reference sequence)

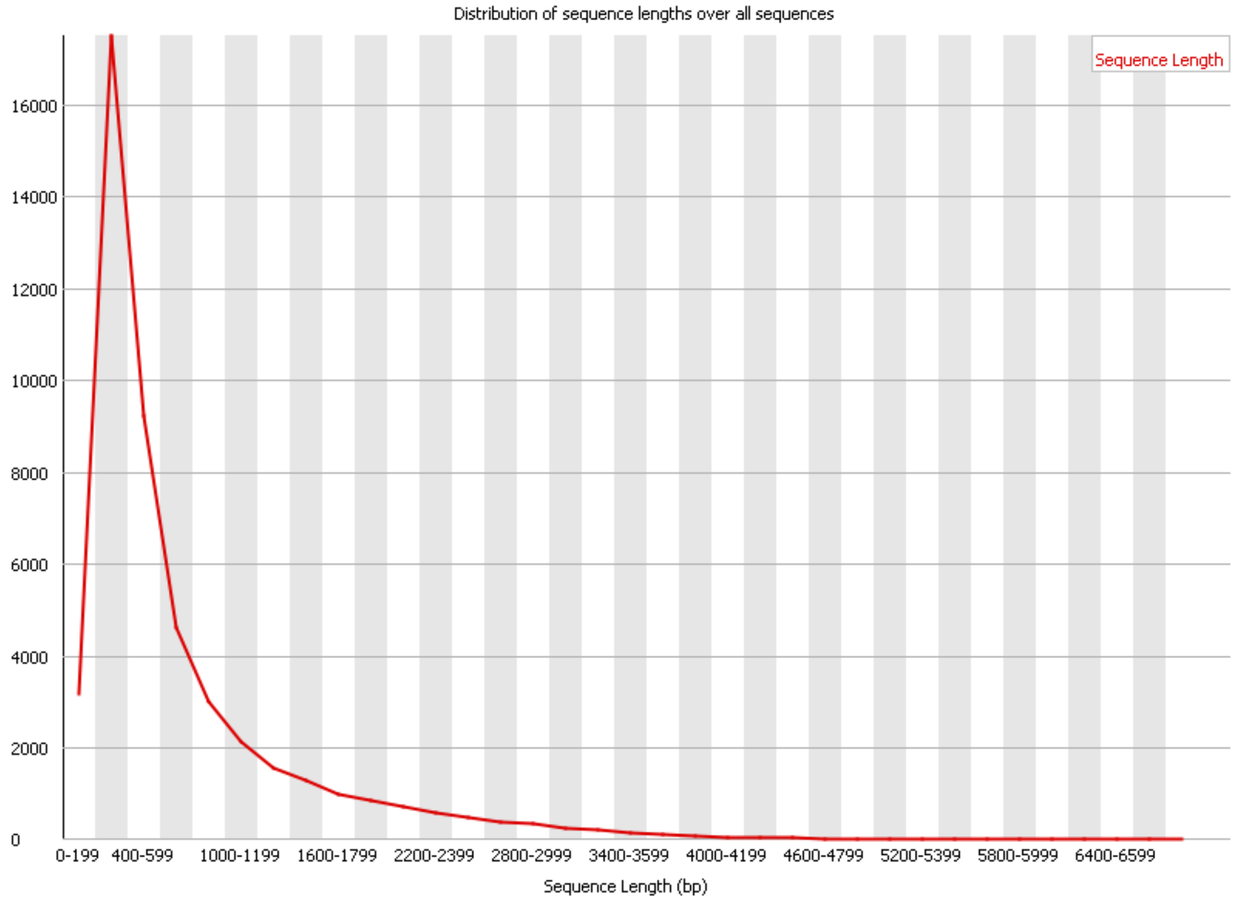


https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

Quality Control: FASTQ

Sequence Length Distribution





https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

Quality Control: FASTQ

Per Base Sequence Content

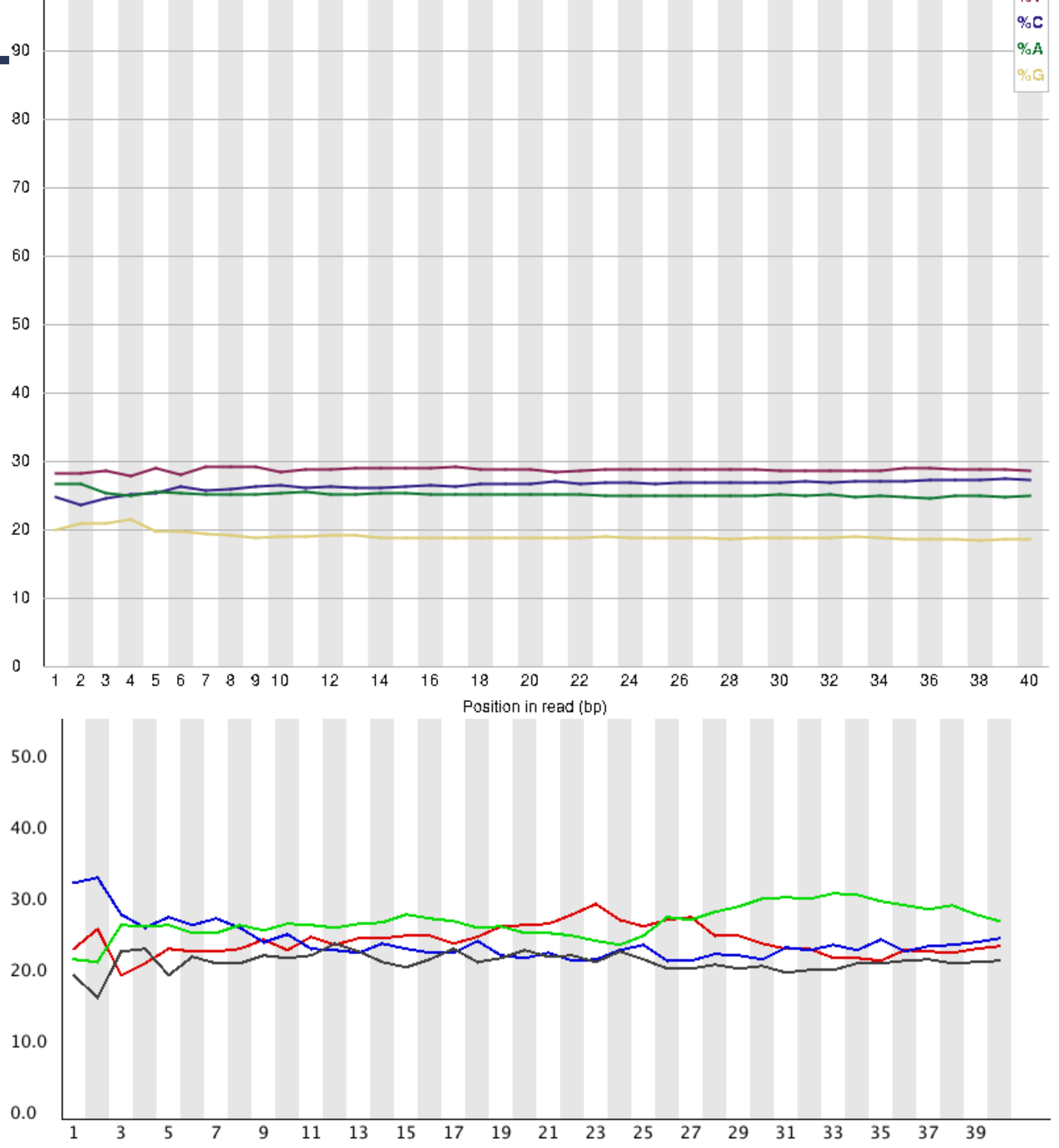
In a random library you would expect that there would be little to no difference between the different bases of a sequence run, so the lines in this plot should run parallel with each other.

There are a number of common scenarios which would ellicit a warning or error from this module.

1. Overrepresented sequences: If there is any evidence of overrepresented sequences such as adapter dimers or rRNA in a sample.

2. Biased fragmentation: This is due to a biased selection of random primers, but doesn't represent any individually biased sequences.

3. Biased composition libraries: Some libraries are inherently biased in their sequence composition.

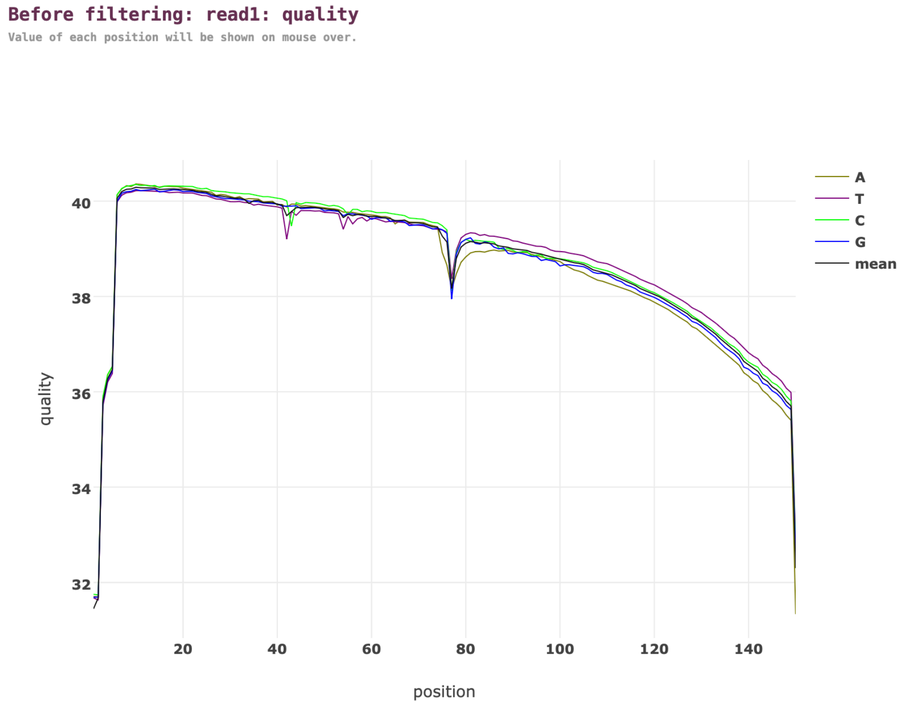


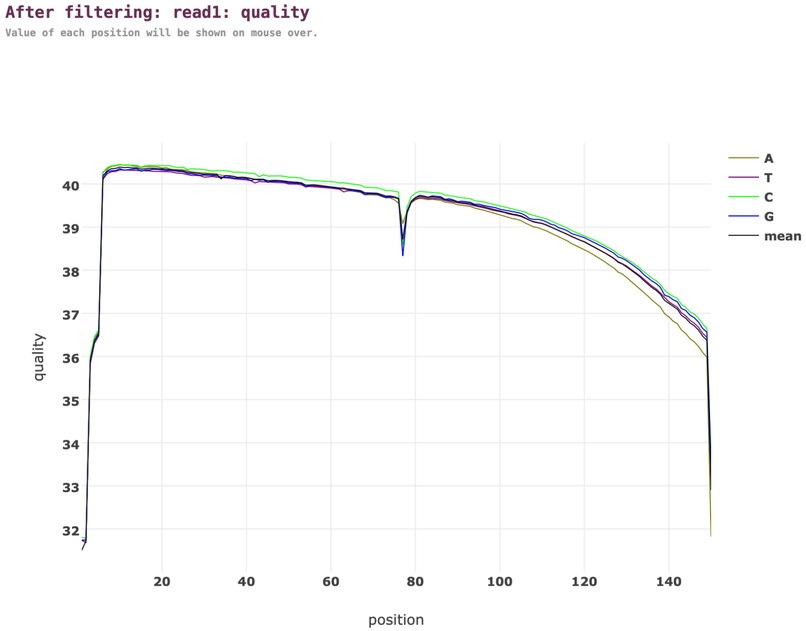
https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

Quality Control: FASTQ

Read Trimming & Filtering with fastp

Operation	What it removes
Adapter trimming	Illumina adapter sequences
Quality trimming	Low Phred-score bases (3' and 5' ends)
Per-base quality filtering	Bases below threshold
Length filtering	Reads too short after trimming
PolyG trimming	NovaSeq-specific artifact
Low complexity filtering	e.g. AAAAAAA... reads





https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

How Much Did We Sequence?

Understanding Throughput and Coverage

Throughput

Total amount of data generated by a sequencing run (Gb or number of reads)

Determined before alignment to a reference genome

Example: NovaSeq X Plus → up to ~16 Tb per run

Coverage (Depth)

Average number of times each base of the genome has been sequenced

$$\text{Coverage} = \frac{\text{Number of reads} \times \text{Read length}}{\text{Genome size}}$$

Example:

Paired end

(15M R1 reads + 15M R2 reads) × 150 bp / 3 Gb = 15×

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

Looking at FASTQ files

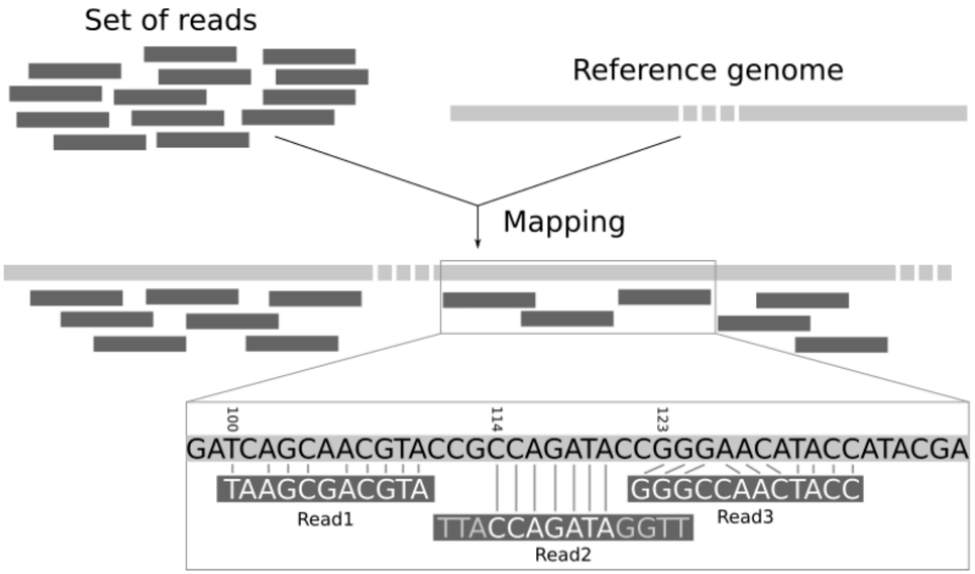
Hands on analysis on Orfeo



Mapping and alignment

Where did this read come from?

Mapping consists in finding the best match between a query sequence and a reference.



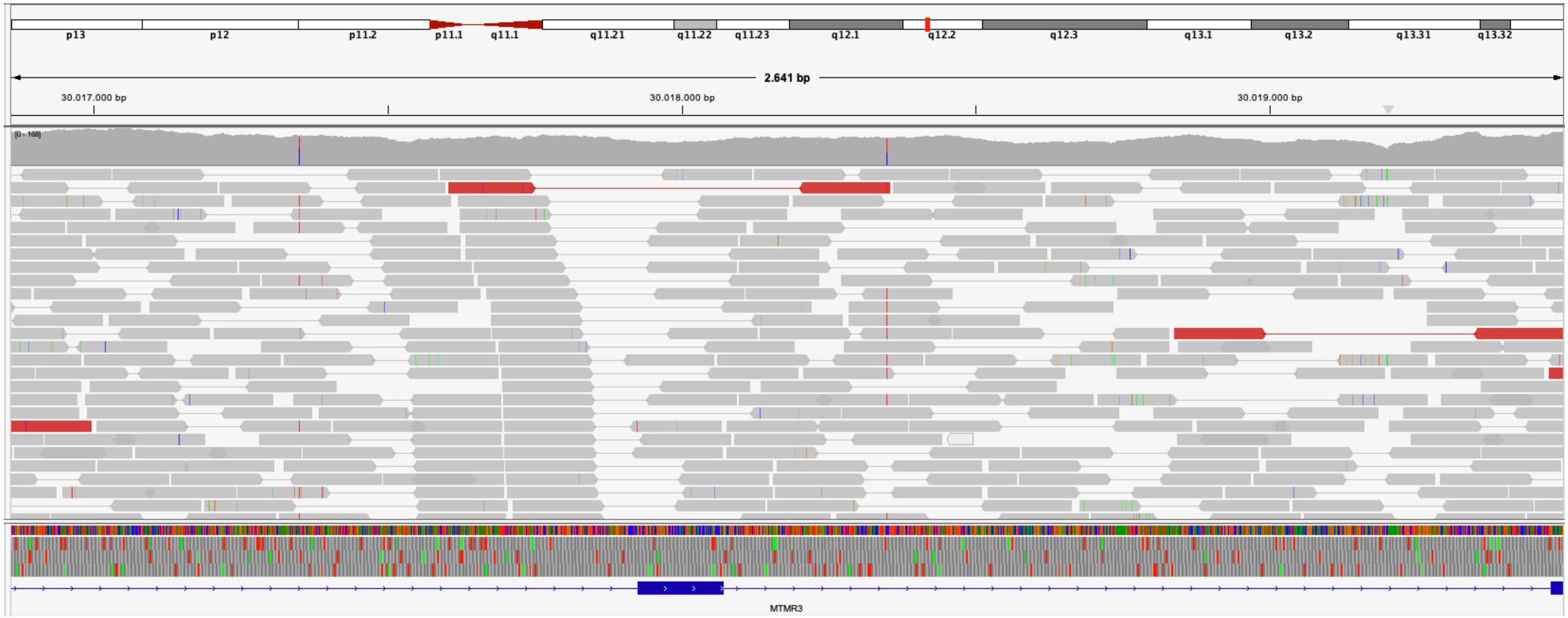
https://training.galaxyproject.org/training-material/topics/sequence-analysis/tutorials/mapping/tutorial_IT.html

Challenges

- The mapper must be able to find inexact alignments by tolerating differences between the reads and the reference.
- Human genome contains repetitive regions.

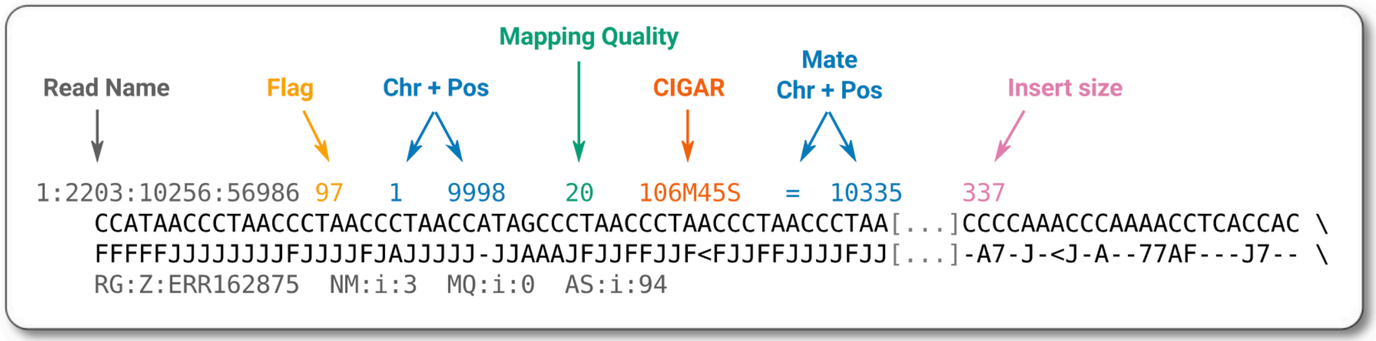
Mapping and alignment

Visualising alignment in IGV



Aligned Sequencing Data: SAM

Reading a SAM record



NGS data formats and Quality Control - Jacqui Keane and Petr Danecek

Field	Meaning
QNAME	Read identifier
FLAG	Bit enocidng the alignment property
RNAME	Chromsome name
POS	Position
MAPQ	Phred-scaled mapping quality probability
CIGAR	Alignment descriptor encoding matches, mismatches, indels, clips
RNEXT	Chromosme name of the mate read
PNEXT	Position of the mate read
TLEN	Insert size length
SEQ	Read sequence
QUAL	Base quality Phred-score

Aligned Sequencing Data: SAM

CIGAR



Compact representation of sequence alignment

M	alignment match or mismatch
=	sequence match
X	sequence mismatch
I	insertion to the reference
D	deletion from the reference
S	soft clipping (clipped sequences present in SEQ)
H	hard clipping (clipped sequences NOT present in SEQ)
N	skipped region from the reference
P	padding (silent deletion from padded reference)

Examples:

```
Ref:   ACGTACGTACGTACGT
Read:  ACGT----ACGTACGA
Cigar: 4M 4D 8M

Ref:   ACGT----ACGTACGT
Read:  ACGTACGTACGTACGT
Cigar: 4M 4I 8M

Ref:   ACTCAGTG--GT
Read:  ACGCA-TGCAGTtagacgt
Cigar: 5M 1D 2M 2I 2M 7S
```

Aligned Sequencing Data: SAM

FLAG and MAPQ

Bit		Description
1	0x1	template having multiple segments in sequencing
2	0x2	each segment properly aligned according to the aligner
4	0x4	segment unmapped
8	0x8	next segment in the template unmapped
16	0x10	SEQ being reverse complemented
32	0x20	SEQ of the next segment in the template being reverse complemented
64	0x40	the first segment in the template
128	0x80	the last segment in the template
256	0x100	secondary alignment
512	0x200	not passing filters, such as platform/vendor quality controls
1024	0x400	PCR or optical duplicate
2048	0x800	supplementary alignment

Pair read analysis

In a chromosome of a parasite genome

	Flag 1	Flag 2	Count	%	Average	Median	STD	Min	Max
← ←	65	129	4	0.000	278849	289087	262174.88	74557	703194
← ←	67	131	4	0.000	109	97	59.98	71	210
← →	81	161	224	0.001	18534.46	53	90016.41	28	1005063
← →	83	163	542	0.003	77.74	65	53.13	4	293
→ ←	97	145	1789	0.009	2320.61	410	29877.06	30	680974
→ ←	**99**	**147**	**99481**	**0.482**	**275.29**	**295**	**79.71**	**61**	**401**
→ →	113	177	7	0.000	306645.43	299601	182414.84	189196	681374
→ →	115	179	4	0.000	141.25	203	98.6	102	259
← ←	129	65	10	0.000	278402.3	237121	198856.09	128485	656117
→ →	131	67	6	0.000	194.67	178	79.93	137	321
← →	145	97	773	0.004	5837.39	52	52533.28	15	903807
← →	147	99	1128	0.005	73.06	68	34.43	4	286
→ ←	161	81	2286	0.011	1823.95	407	21527.69	15	597483
→ ←	**163**	**83**	**100010**	**0.485**	**273.92**	**295**	**80.98**	**59**	**401**
← ←	177	113	7	0.000	170902.43	102523	149875.07	44144	431897
← ←	179	115	12	0.000	221	255	108.48	92	378

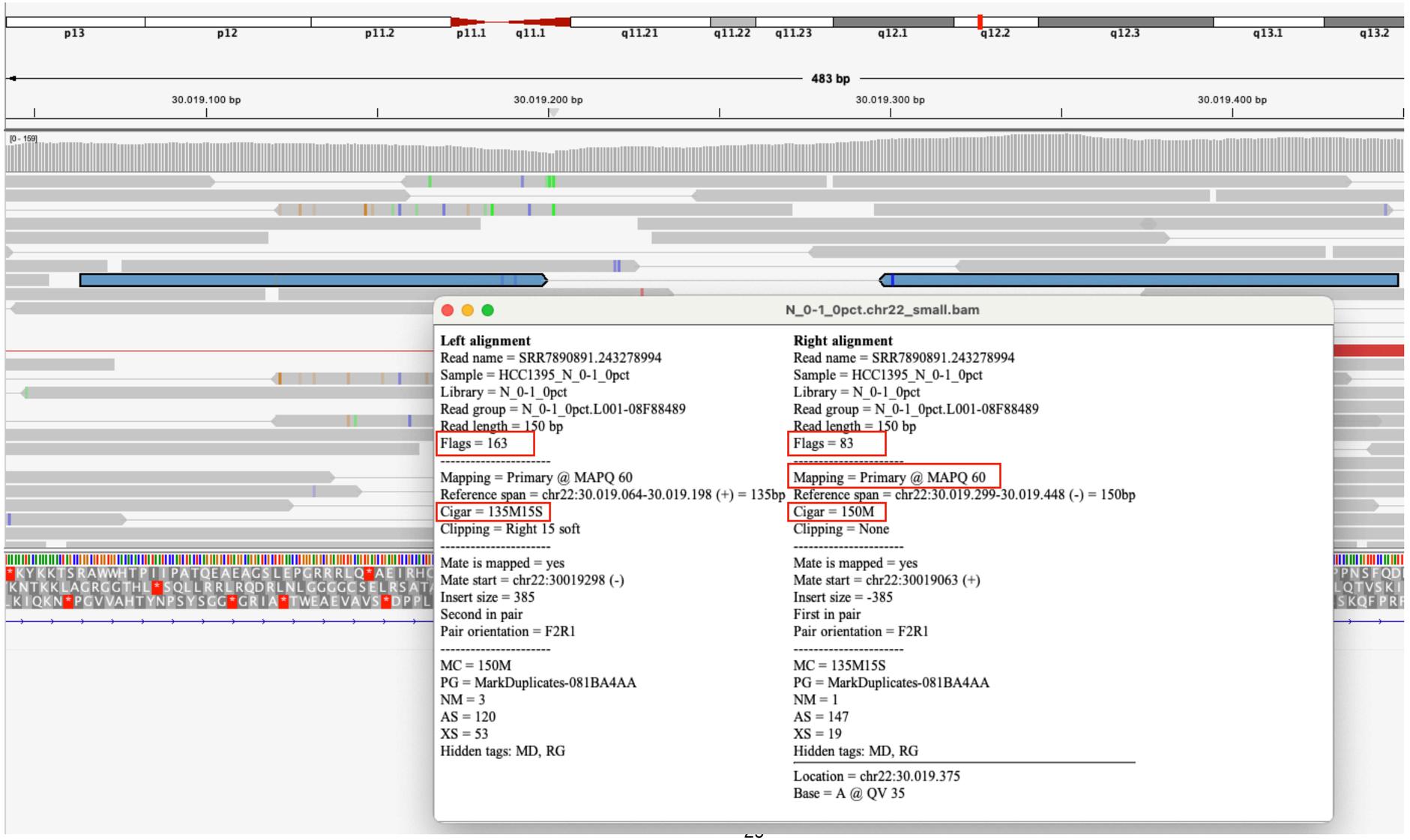
Only 99, 147 and 163, 83 are properly mapped read pairs within a defined insert size
Single reads are not shown

$$MAPQ = -10 \times \log_{10}(P_{wrongmapping})$$

- **MAPQ= 60** $\rightarrow \frac{1}{1000000}$ of placing the read in the wrong position
- **MAPQ=0** → multiple equally-good alignment positions (there are regions hard to be mapped)

Aligned Sequencing Data: SAM

FLAG and MAPQ - visualising on IGV



FLAG 163 =
read paired (0x1)
read mapped in proper pair (0x2)
mate reverse strand (0x20)
second in pair (0x80)

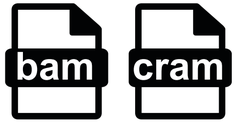
FLAG 83 =
read paired (0x1)
read mapped in proper pair (0x2)
read reverse strand (0x10)
first in pair (0x40)

CIGAR 135M15S =
135 matching bases
5 substituted bases

CIGAR 150M =
150 matching bases

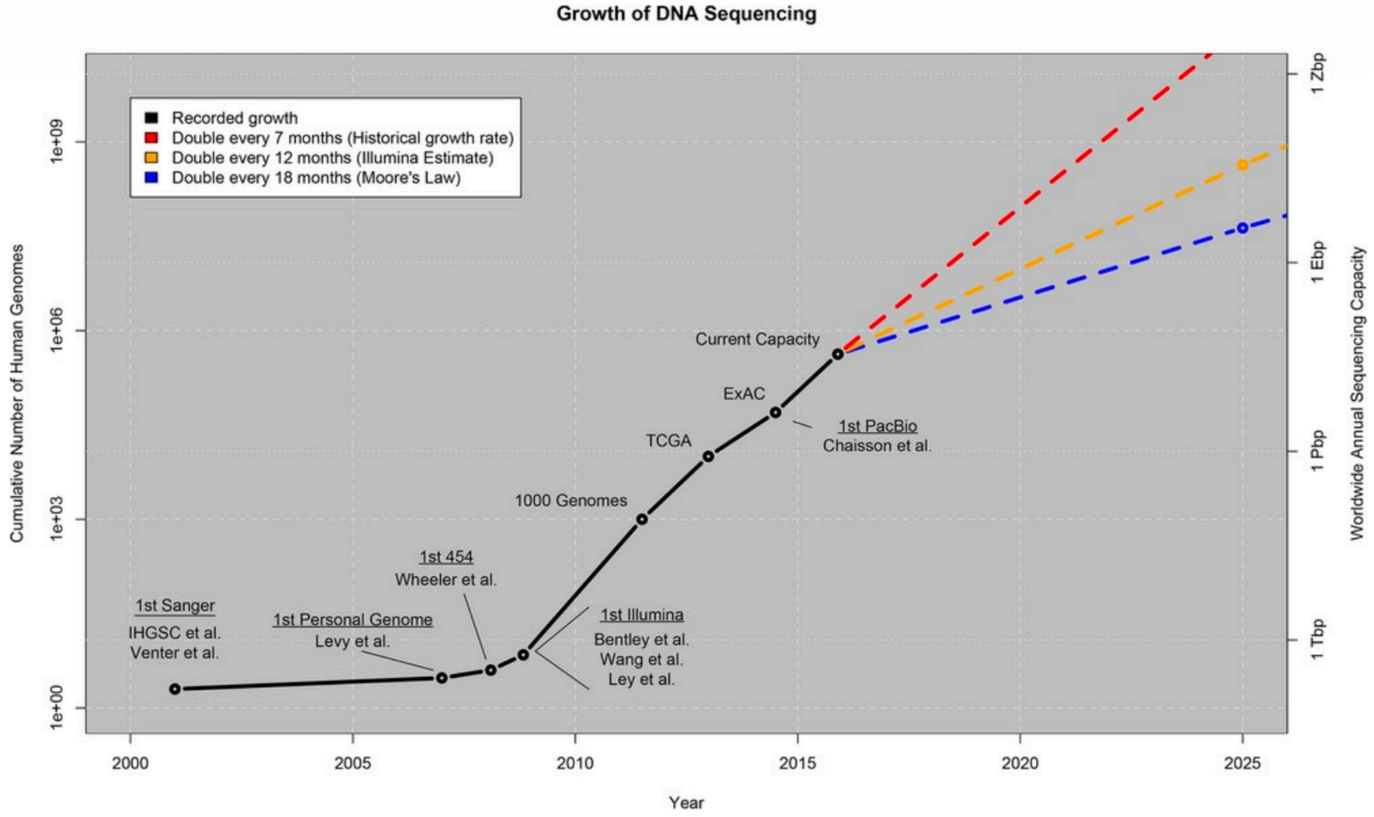
Aligned Sequencing Data: BAM and CRAM

Compressing alignment files



BAM files are too large
- ~1.5-2 bytes per base pair
- Increases in disk capacity are being far outstripped by sequencing technologies

CRAM: in lossless mode 60% of BAM size
- Reference based compression
- Controlled loss of quality information



Stephens, Zachary D., et al. "Big data: astronomical or genomical?." *PLoS biology* 13.7 (2015): e1002195.

Quality Control: BAM

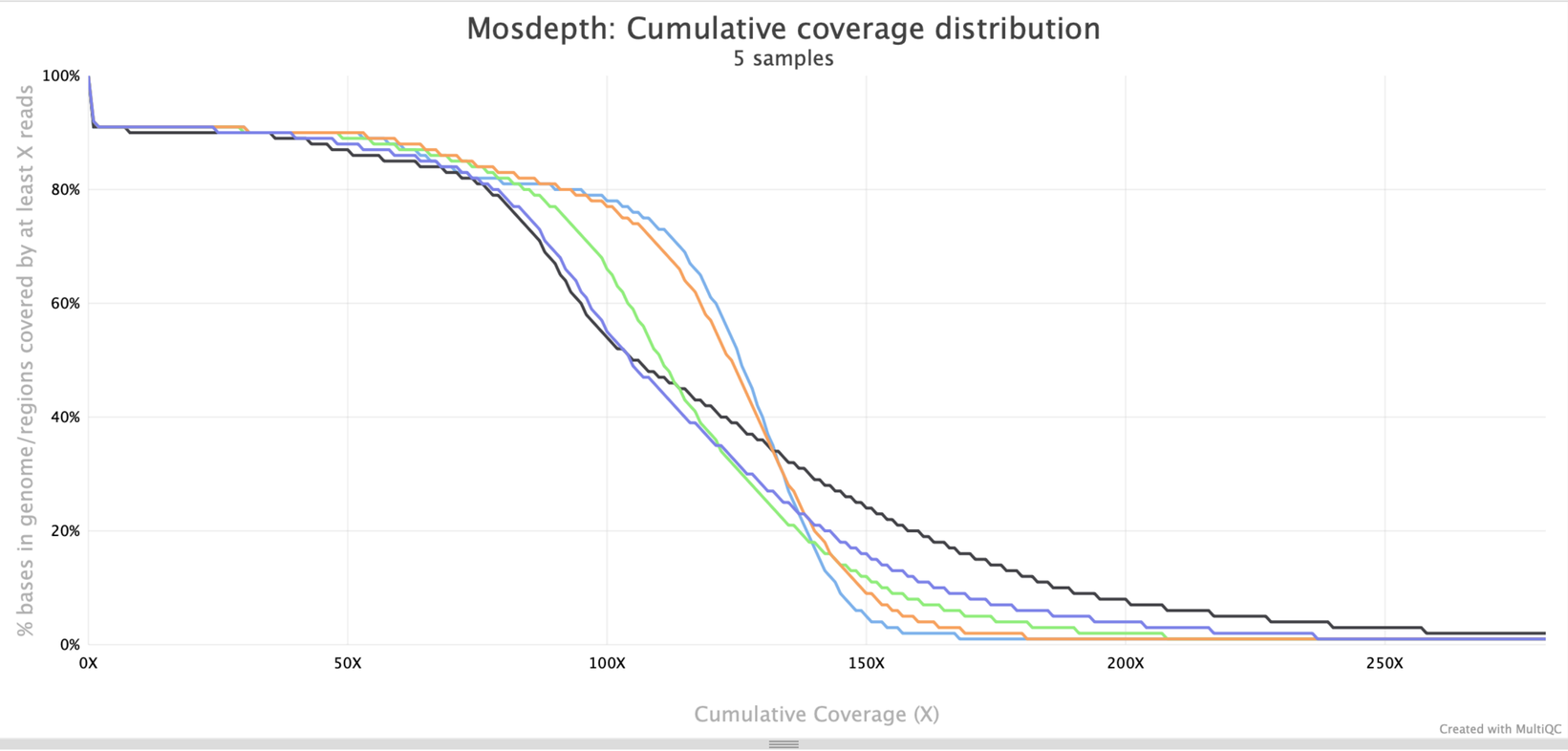
Coverage

Cumulative coverage distribution

Help

Proportion of bases in the reference genome with, at least, a given depth of coverage. Note that for 5 samples, a BED file was provided, so the data was calculated across those regions. For 5 samples, it's calculated across the entire genome length. 5 samples have both global and region reports, and we are showing the data for regions

Export Plot



Breadth of Coverage

Fraction of the reference genome covered by at least one read (or ≥ a minimum depth, e.g. ≥10×)

Expressed as a percentage

Example: 90% breadth at ≥10× → 10% of the genome poorly/not covered

Quality Control: BAM

Mapping statistics

Samtools Flagstat

Toolkit for interacting with BAM/CRAM files. *URL: http://www.htslib.org DOI: 10.1093/bioinformatics/btp352*

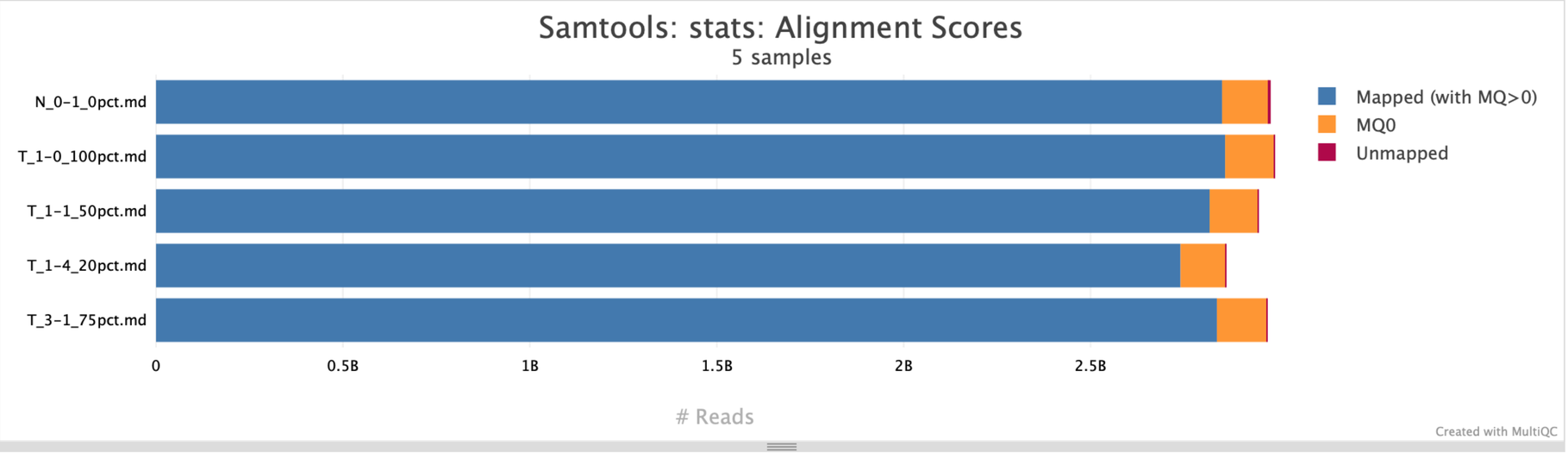
Percent mapped

Help

Alignment metrics from `samtools stats`; mapped vs. unmapped reads vs. reads mapped with MQ0.

Percentages

Export Plot



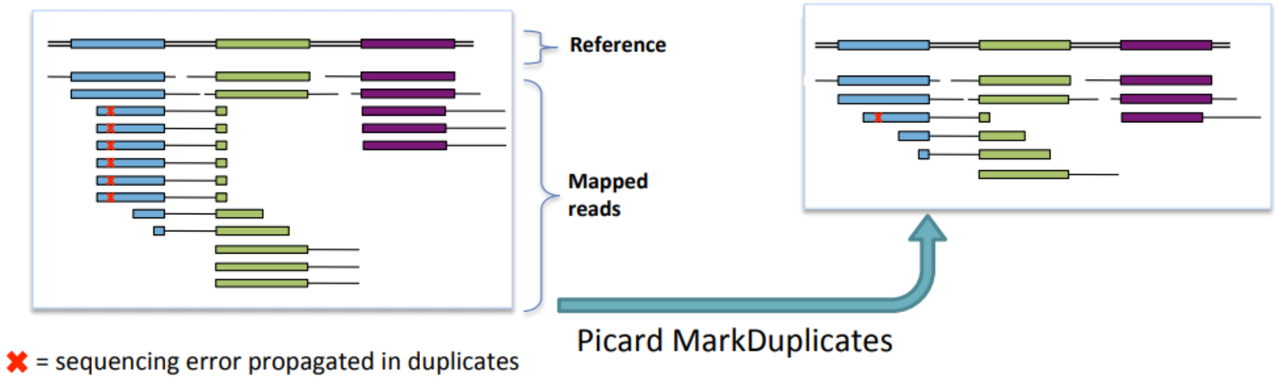
Post-processing BAM

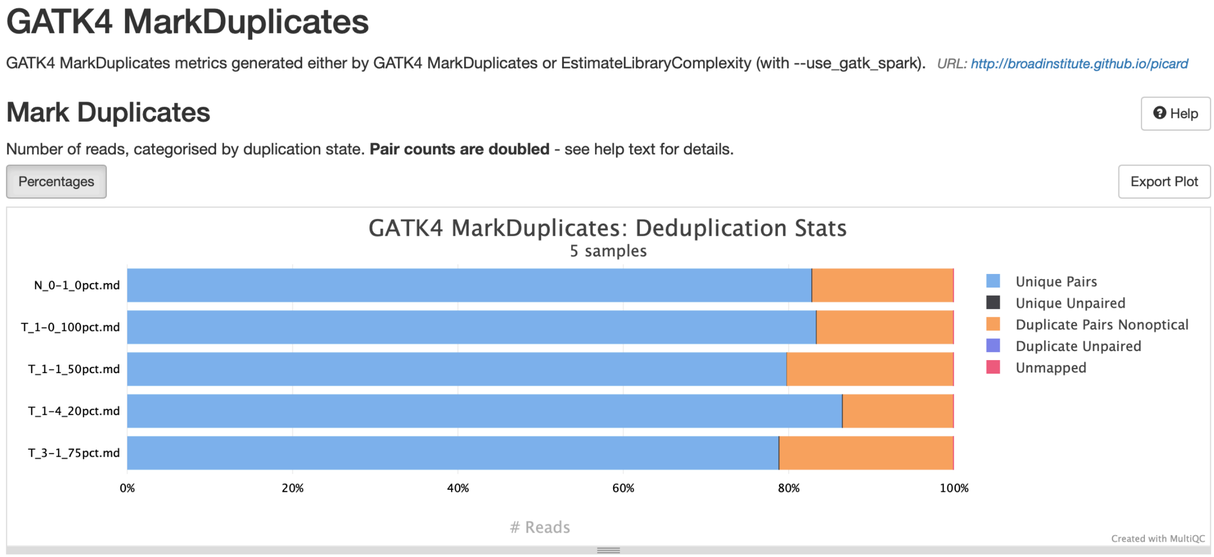
Duplicate Rate and Markduplicates

A final step in the mapping pipeline is the tagging (or removal) of PCR duplicates.

These are identical sequences that can arise because of PCR steps during library preparation or because a cluster on the flow cell was erroneously detected as two separate one.

The overall goal of PCR duplicate removal is to reduce bias during downstream analyses (like variant detection) due to their presence.



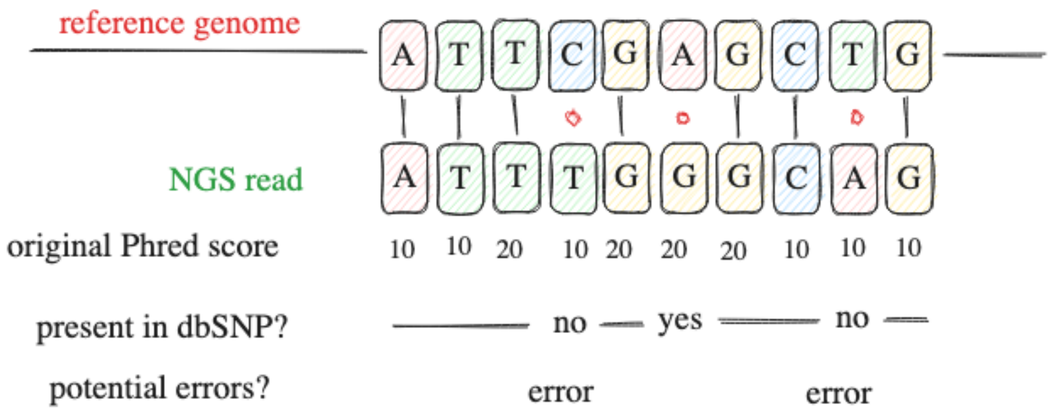


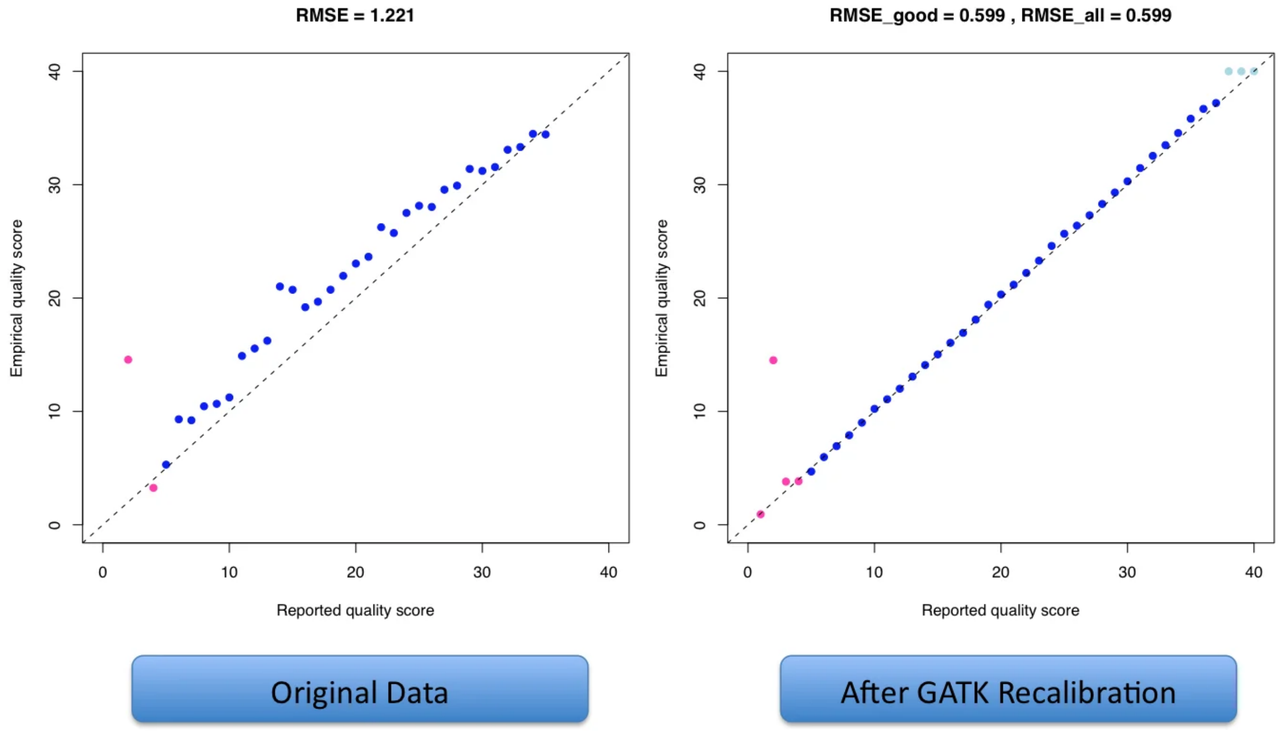
A beginners guide to Call SNPs and indels - Lijia Yu

Post-processing BAM

BQSR - Base Quality Score Recalibration

Once a read has been aligned to the reference, an appropriate algorithm can however compare the error rate estimated from the existing base quality scores, with the actual differences observed with the reference sequence (empirical quality), and perform appropriate corrections.



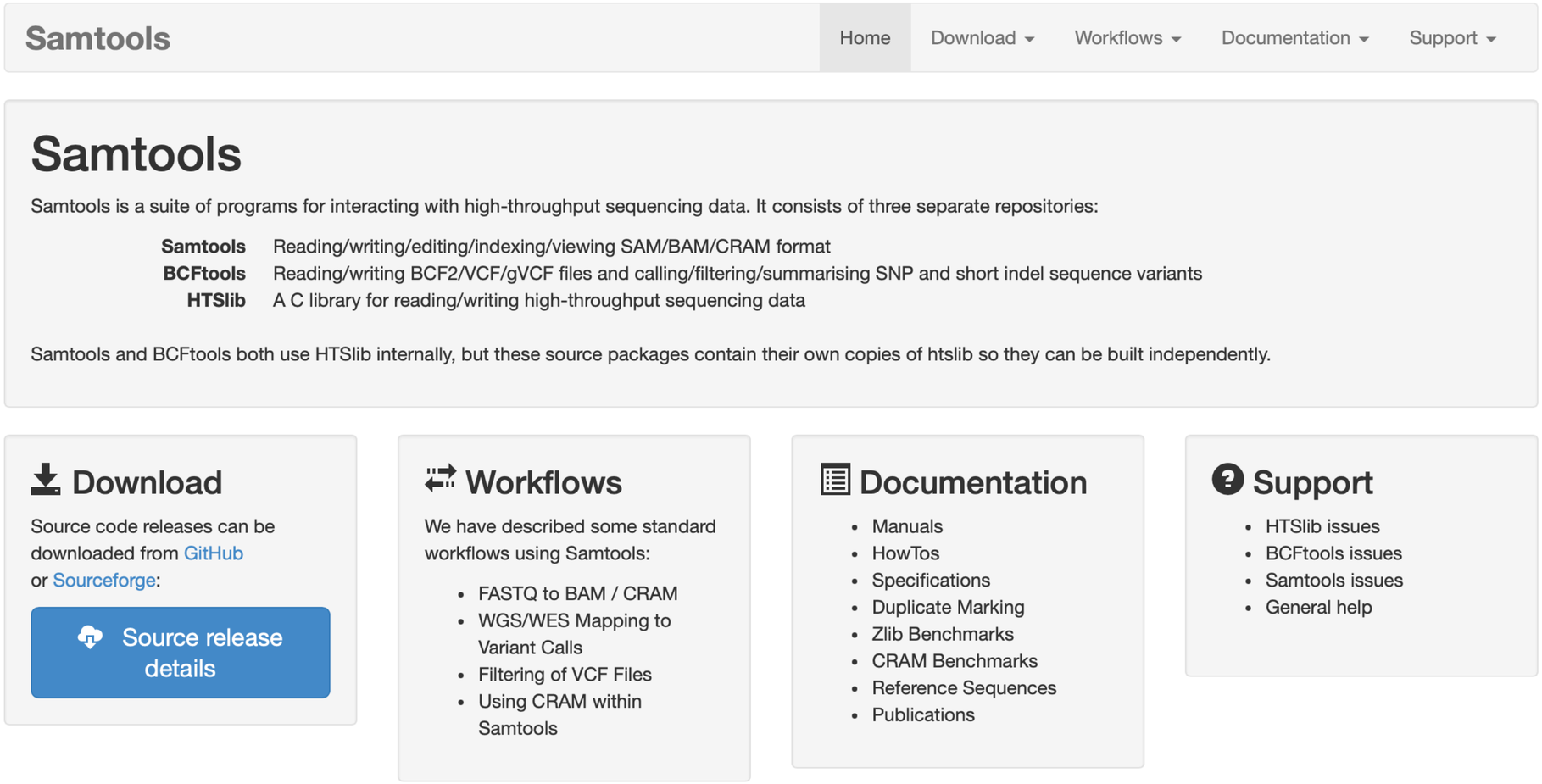


Original Data

After GATK Recalibration

Looking at BAM files

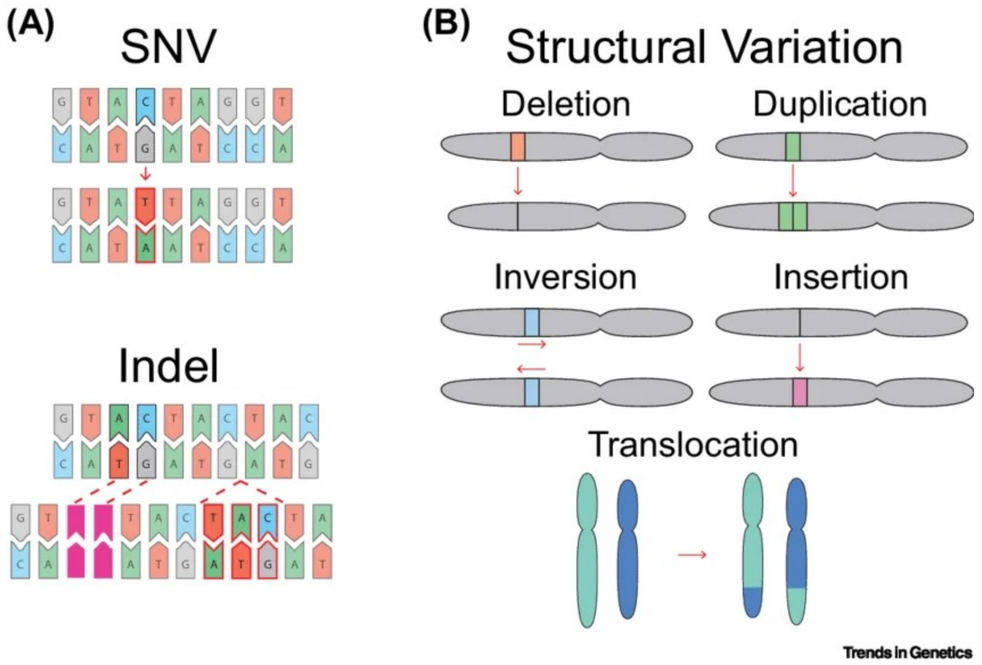
How to use samtools



What’s next?

Variant Calling

Variant calling aims at identifying single nucleotide variants (SNVs), small insertions and deletion (indels), copy number alterations (CNA) and structural variants (SV) from next generation sequencing data.



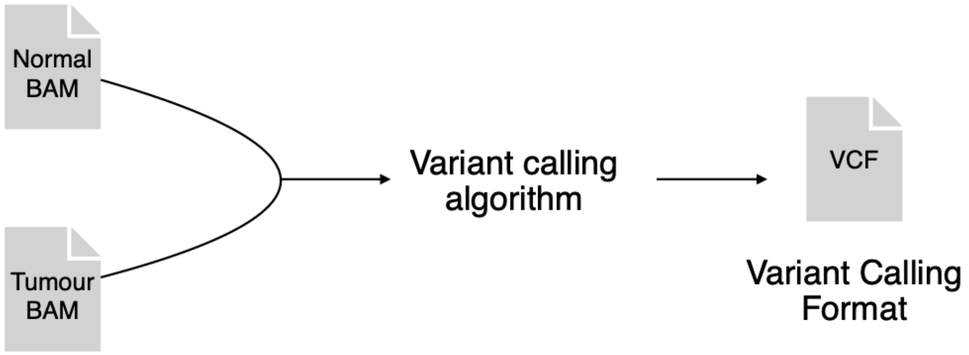
Nesta, A. V., Tafur, D., & Beck, C. R. (2021). Hotspots of human mutation. *Trends in Genetics*, *37*(8), 717-729.

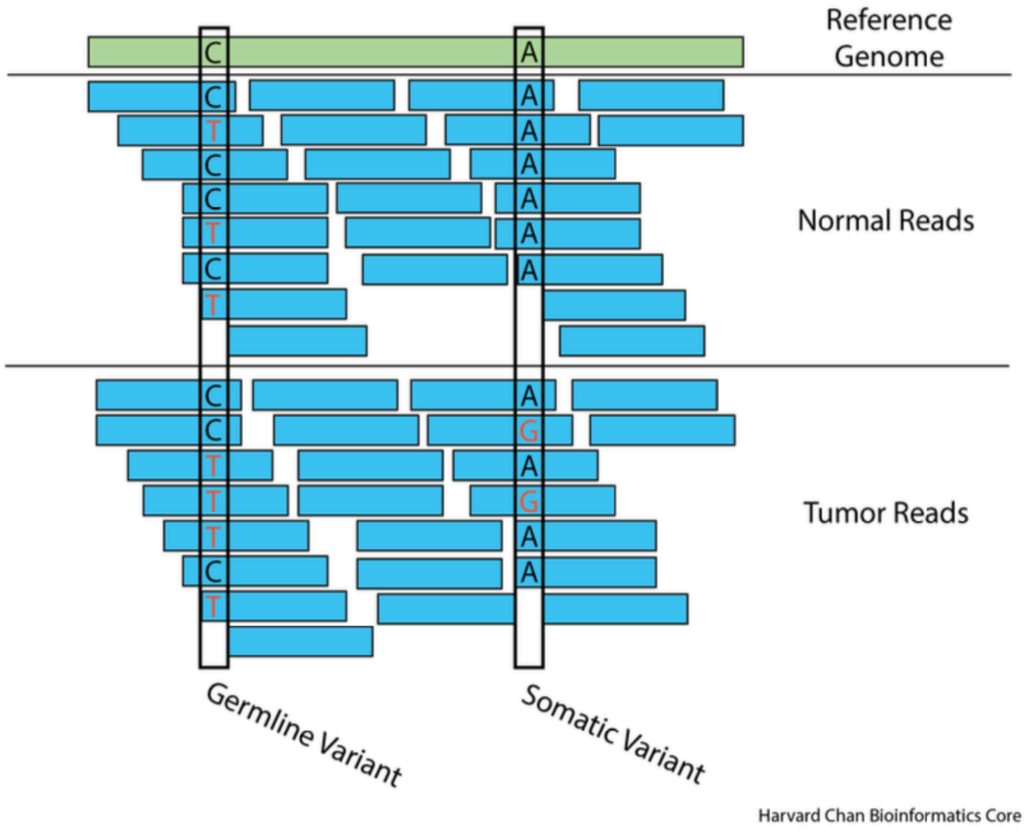
Germline vs Somatic

Variant Calling

Germline mutations are changes to your DNA that you inherit from the egg and sperm cells and it is passes directly from a parent to a child.
All the cells of the body inherit germline mutations.

Somatic mutations are changes to DNA that happens after conception, are not present in every cell in the body, and are not passed from parent to child.
Somatic mutations represent the major causes of cancer.





VCF files

How a variant call is recorded



VCF header

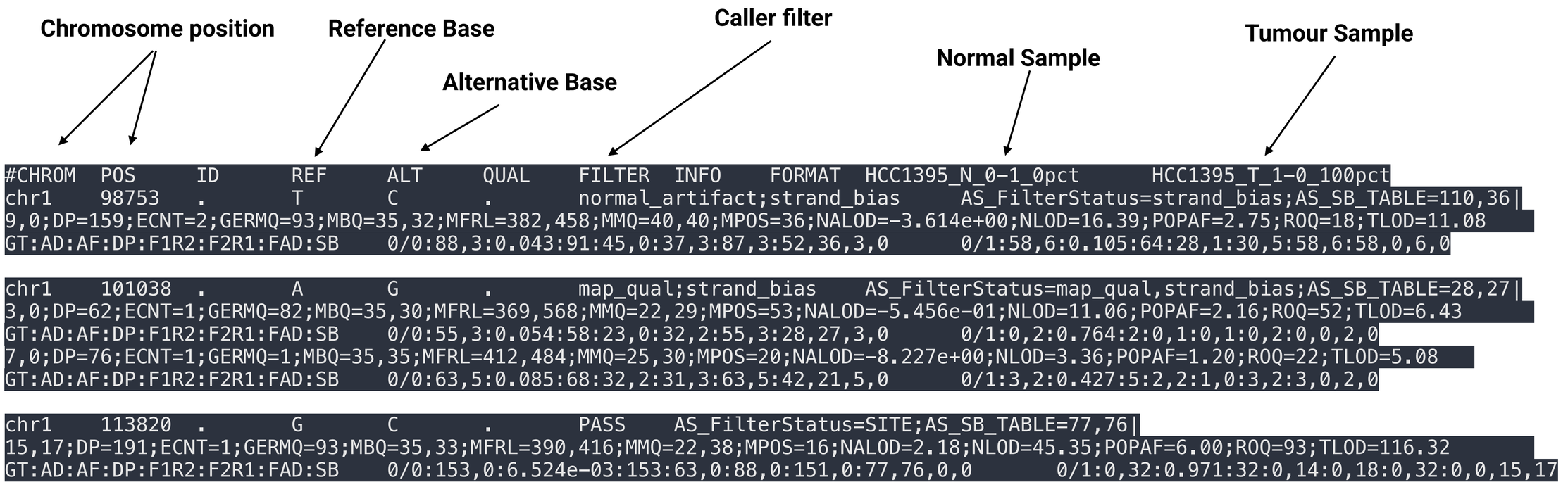
```
##fileformat=VCFv4.2
##FILTER=<ID=FAIL,Description="Fail the site if all alleles fail but for different reasons.">
##FILTER=<ID=PASS,Description="Site contains at least one allele that passes filters">
##FORMAT=<ID=AD,Number=R,Type=Integer,Description="Allelic depths for the ref and alt alleles in the order listed">
##FORMAT=<ID=AF,Number=A,Type=Float,Description="Allele fractions of alternate alleles in the tumor">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth (reads with MQ=255 or with bad mates are filtered)">
##FORMAT=<ID=F1R2,Number=R,Type=Integer,Description="Count of reads in F1R2 pair orientation supporting each allele">
##FORMAT=<ID=F2R1,Number=R,Type=Integer,Description="Count of reads in F2R1 pair orientation supporting each allele">
##FORMAT=<ID=FAD,Number=R,Type=Integer,Description="Count of fragments supporting each allele.">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=PGT,Number=1,Type=String,Description="Physical phasing haplotype information, describing how the alternate alleles
are phased in relation to one another; will always be heterozygous and is not intended to describe called alleles”>
```

VCF body

```
#CHROM  POS     ID      REF     ALT     QUAL    FILTER  INFO    FORMAT  HCC1395_N_0-1_0pct      HCC1395_T_1-4_20pct
chr1    14699   .       C       G       .       normal_artifact;strand_bias     AS_FilterStatus=strand_bias;AS_SB_TABLE=44,43|
0,10;DP=101;ECNT=1;GERMQ=77;MBQ=27,35;MFRL=382,460;MMQ=25,30;MPOS=49;NALOD=-3.116e+00;NLOD=12.87;POPAF=0.428;ROQ=89;TLOD=18.65
GT:AD:AF:DP:F1R2:F2R1:FAD:SB    0/0:74,3:0.051:77:35,1:34,2:73,3:40,34,0,3      0/1:13,7:0.400:20:5,3:6,4:11,7:4,9,0,7
chr1    16977   .       G       A       .       map_qual        AS_FilterStatus=map_qual;AS_SB_TABLE=74,48|
4,7;DP=134;ECNT=1;GERMQ=13;MBQ=35,34;MFRL=365,593;MMQ=23,22;MPOS=38;NALOD=-1.757e+01;NLOD=7.52;POPAF=0.400;ROQ=67;TLOD=8.28
GT:AD:AF:DP:F1R2:F2R1:FAD:SB    0/0:120,7:0.062:127:49,4:66,3:120,7:72,48,0,7   0/1:2,4:0.624:6:2,2:0,2:2,4:2,0,4,0
```

VCF files

How a variant call is recorded



Format of genotype field

GT = genotype
AD = allele depth
AF = allele fraction of ALT
DP = read depth

Genotype information of normal sample

GT = 0/0
AD = 153,0
AF = 6.524e-03
DP = 153

Genotype information of tumour sample

GT = 0/1
AD = 0,32
AF = 0.971
DP = 32

→ Reads covering only the alternative base (C)
Lower coverage

BED files

Variant Calling



- BED (Browser Extensible Data) format provides a flexible way to define the data lines that are displayed in an annotation track.
- BED lines have three required fields and nine additional optional fields.
- BED identifies a region you want to inspect or analyze.

The first three required BED fields are:

1. **chrom** - The name of the chromosome (e.g. chr3, chrY, chr2_random) or scaffold (e.g. scaffold10671).

2. **chromStart** - The starting position of the feature in the chromosome or scaffold. The first base in a chromosome is numbered 0.

4. **chromEnd** - The ending position of the feature in the chromosome or scaffold. The *chromEnd* base is not included in the display of the feature, however, the number in **position format** will be represented. For example, the first 100 bases of chromosome 1 are defined as *chrom=1, chromStart=0, chromEnd=100*, and span the bases numbered 0-99 in our software (not 0-100), but will represent the position notation chr1:1-100.

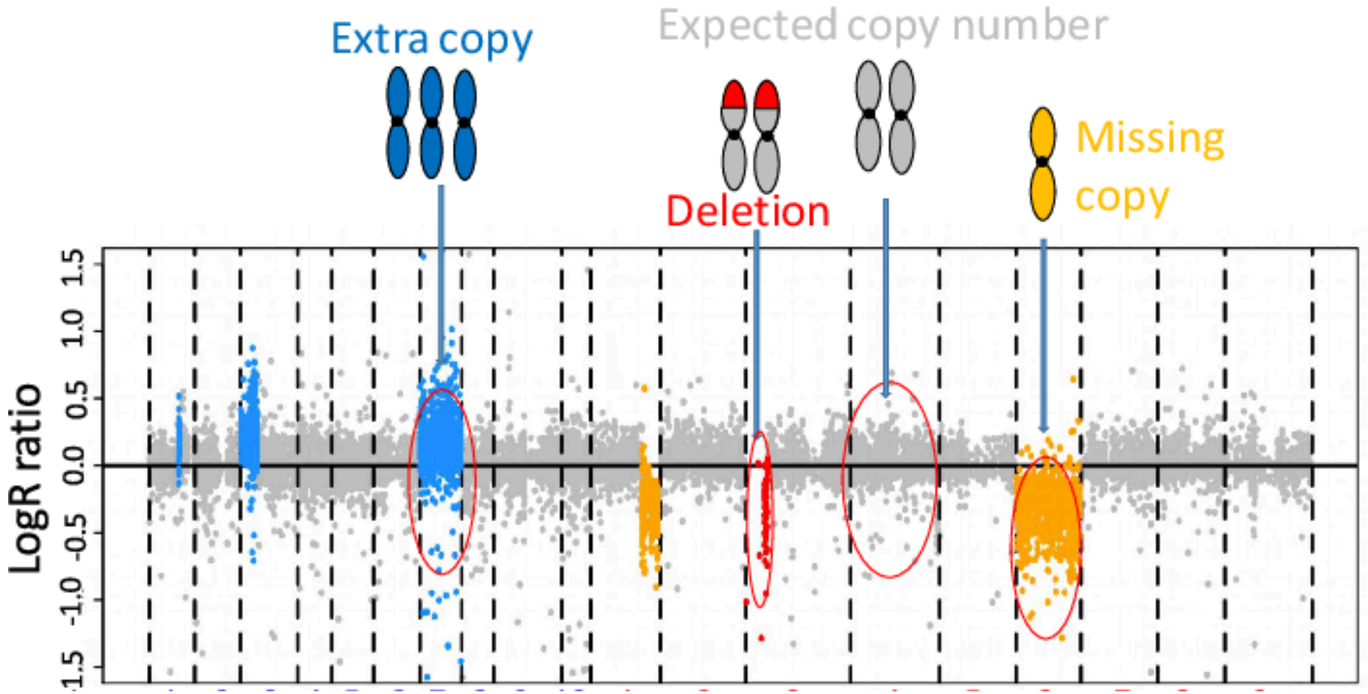
Copy Number calling

Variant Calling

Principles

- More DNA copies → more reads aligning to that region

- Read depth (coverage) is normalized against a matched normal and $\log_2$-transformed: $\log_2$(tumor/normal)



Katche, Elizabeth Ihien, et al. "Genetic factors inherited from both diploid parents interact to affect genome stability and fertility in resynthesized allotetraploid Brassica napus." *G3: Genes, Genomes, Genetics* 13.8 (2023):

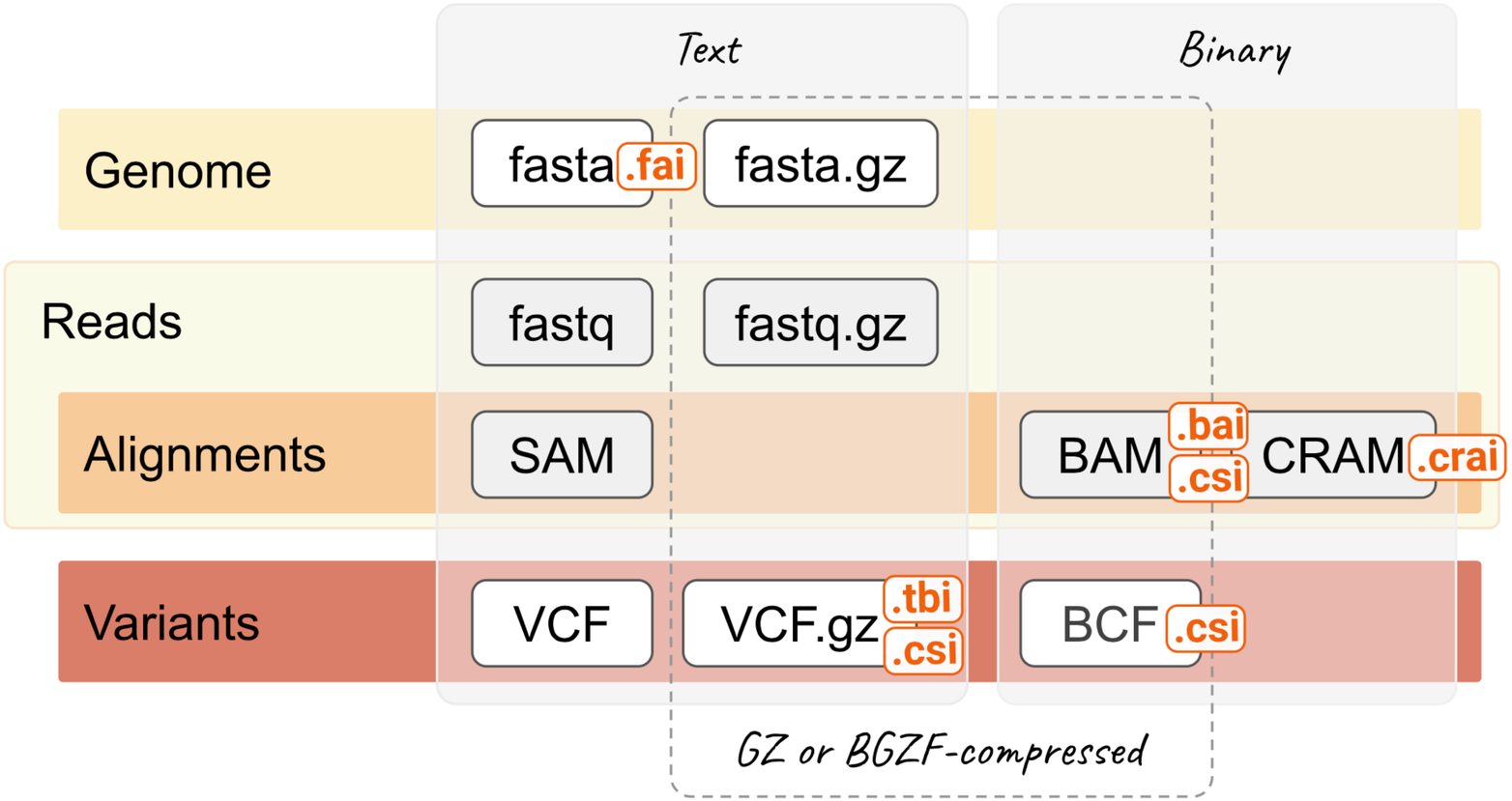
Copy Number calling

File format

chromosome	start	end	cn
chr21	5009996	5045716	1
chr21	5045792	5053240	5
chr21	5054380	5108720	1
chr21	5108720	5112900	3
chr21	5112900	5159792	1
chr21	5159792	5160628	5
chr21	5161540	5165568	1
chr21	5165720	5511976	4
chr21	5512052	5519576	3

File format summary

Variant Calling



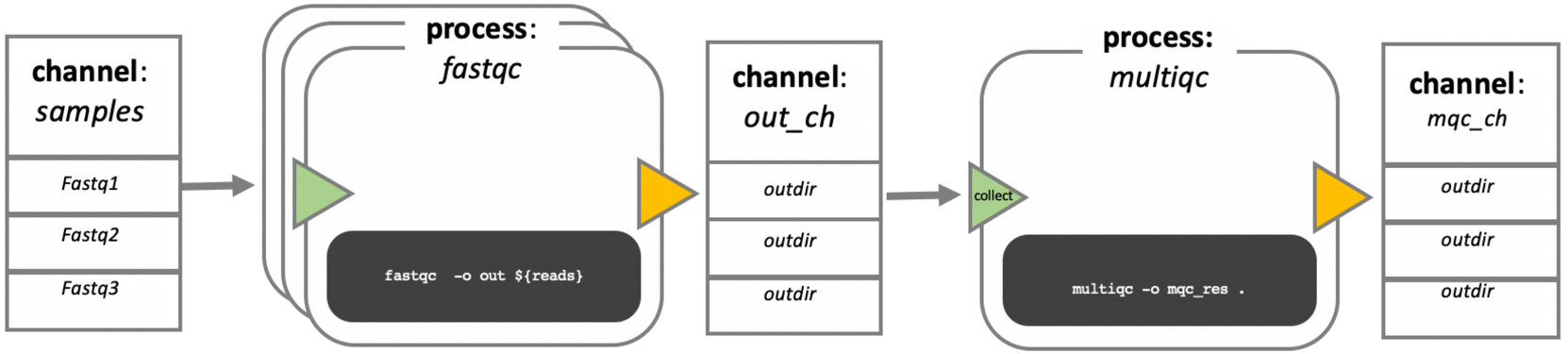
Nextflow pipelines

Nextflow is a workflow system for creating **scalable**, **portable**, and **reproducible** workflows.

It uses a dataflow programming model that simplifies writing parallel and distributed pipelines by allowing you to focus on data flow and computation.

Nextflow can deploy workflows on a variety of execution platforms, including your local machine, HPC schedulers, and cloud.

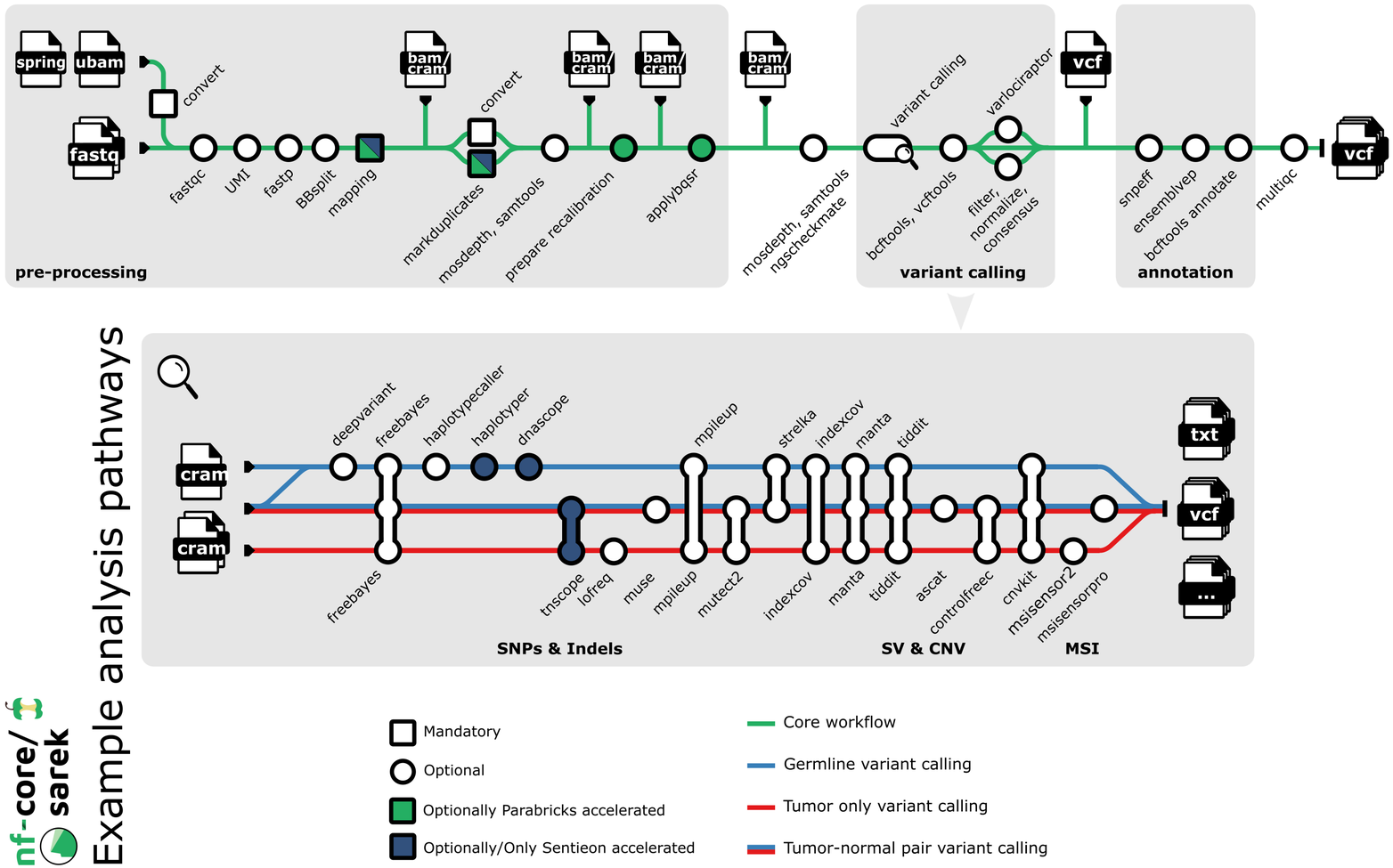
Additionally, Nextflow supports a range of compute environments, software container runtimes, and package managers, allowing workflows to be executed in reproducible and isolated environments.



The Carpentries Incubator

nf-core/sarek

https://nf-co.re/sarek/



Adapted from: Fellows Yates, James A., et al. PeerJ 9 (2021).

Samplesheet

Column	Description
patient	Custom patient ID; designates the patient/subject; must be unique for each patient, but one patient can have multiple samples (e.g. normal and tumor)
sex	Sex chromosomes of the patient; i.e. XX, XY…
status	Normal/tumor status of sample; can be 0 (normal) or 1 (tumor).
sample	Custom sample ID for each tumor and normal sample; more than one tumor sample for each subject is possible. Sample IDs must be unique for unique biological samples
fastq_1	Full path to FastQ file for Illumina short reads 1.
fastq_2	Full path to FastQ file for Illumina short reads 2.
bam	Full path to (u)BAM file
bai	Full path to BAM index file
cram	Full path to CRAM file
crai	Full path to CRAM index file

patient	sex	status	sample	fastq_1	fastq_2
patient1	XX	0	normal_sample	test_1.fastq.gz	test_2.fastq.gz
patient1	XX	1	tumor_sample	test2_1.fastq.gz	test2_2.fastq.gz

Running the pipeline on Orfeo!

```
❯ ssh orfeo

        SCIENCE PARK
   ORFEO HPC · v1.3.0

 HELP
 Docs       https://orfeo-doc.areasciencepark.it
 Support    orfeo.helpdesk@areasciencepark.it

 STORAGE
 HOME     200 GB quota
 SCRATCH  /orfeo/scratch/<group>/<user>
 FAST     /orfeo/fast/<group>/<user>
 LTS      /orfeo/LTS/<group>/LT_Storage
 Home usage: myquota

 JOB TOOLS
 hpcquota           CPU/GPU usage
 lastjobs           recent Job IDs
 seff-rest <JOBID>  resource usage report for a completed job

 JOB SUBMISSION
 Always specify: -A <ACCOUNT>
     sbatch -A <ACCOUNT> job.sh     srun -A <ACCOUNT> ...

 Your Slurm associations (account / partition):
     sacctmgr list associations Users=$(whoami) \
         format=Account,User,Partition

 SECURITY NOTICE
 ORFEO admins will NEVER ask for passwords or SSH private keys.

Last login: Sat Nov  7 16:05:02 2026 from 84.115.211.115
lvaleriani@login02:~$
```

References

Tool	Paper	GitHub	Link
Nextflow	Nextflow enables reproducible	https://github.com/nextflow-io/nextflow	https://www.nextflow.io
nf-core	The nf-core framework for community-	https://github.com/nf-core	https://nf-co.re
nf-core/sarek	Sarek: A portable workflow for	https://github.com/nf-core/sarek	https://nf-co.re/sarek
GATK Best Practices	The Genome Analysis Toolkit: a MapReduce	https://github.com/broadinstitute/gatk	https://gatk.broadinstitute.org
EPI2ME wf-basecalling	-	https://github.com/epi2me-labs/wf-basecalling	-
EPI2ME wf-human-variation	-	https://github.com/epi2me-labs/wf-human-variation	-
FastQC	-	https://github.com/s-andrews/FastQC	https://www.bioinformatics.babraham.ac.uk/projects/fastqc/
fastp	fastp: an ultra-fast all-in-one FASTQ	https://github.com/OpenGene/fastp	-
MultiQC	MultiQC: summarize analysis results	https://github.com/MultiQC/MultiQC	https://multiqc.info
Dorado	-	https://github.com/nanoporetech/dorado	-
BWA-MEM	Aligning sequence reads, clone	https://github.com/lh3/bwa	-
minimap2	Minimap2: pairwise alignment for	https://github.com/lh3/minimap2	-
SAMtools/bcftools	Twelve years of SAMtools and BCFtools	https://github.com/samtools/samtools https://github.com/samtools/bcftools	https://www.htslib.org
HaplotypeCaller	Scaling accurate genetic variant	https://github.com/broadinstitute/gatk	https://gatk.broadinstitute.org
DeepVariant	A universal SNP and small-indel variant	https://github.com/google/deepvariant	-
Mutect2	Calling somatic SNVs and indels with	https://github.com/broadinstitute/gatk	https://gatk.broadinstitute.org
Strelka2	Strelka2: fast and accurate calling of	https://github.com/Illumina/strelka	-
Clair3	Symphonizing pileup and full-alignment for	https://github.com/HKU-BAL/Clair3	-
IGV	Integrative genomics viewer	https://github.com/igvteam/igv	https://igv.org